



RACK-SCALE ARCHITECTURE

FOR WAFER SCALE ENGINE

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Cerebras Systems

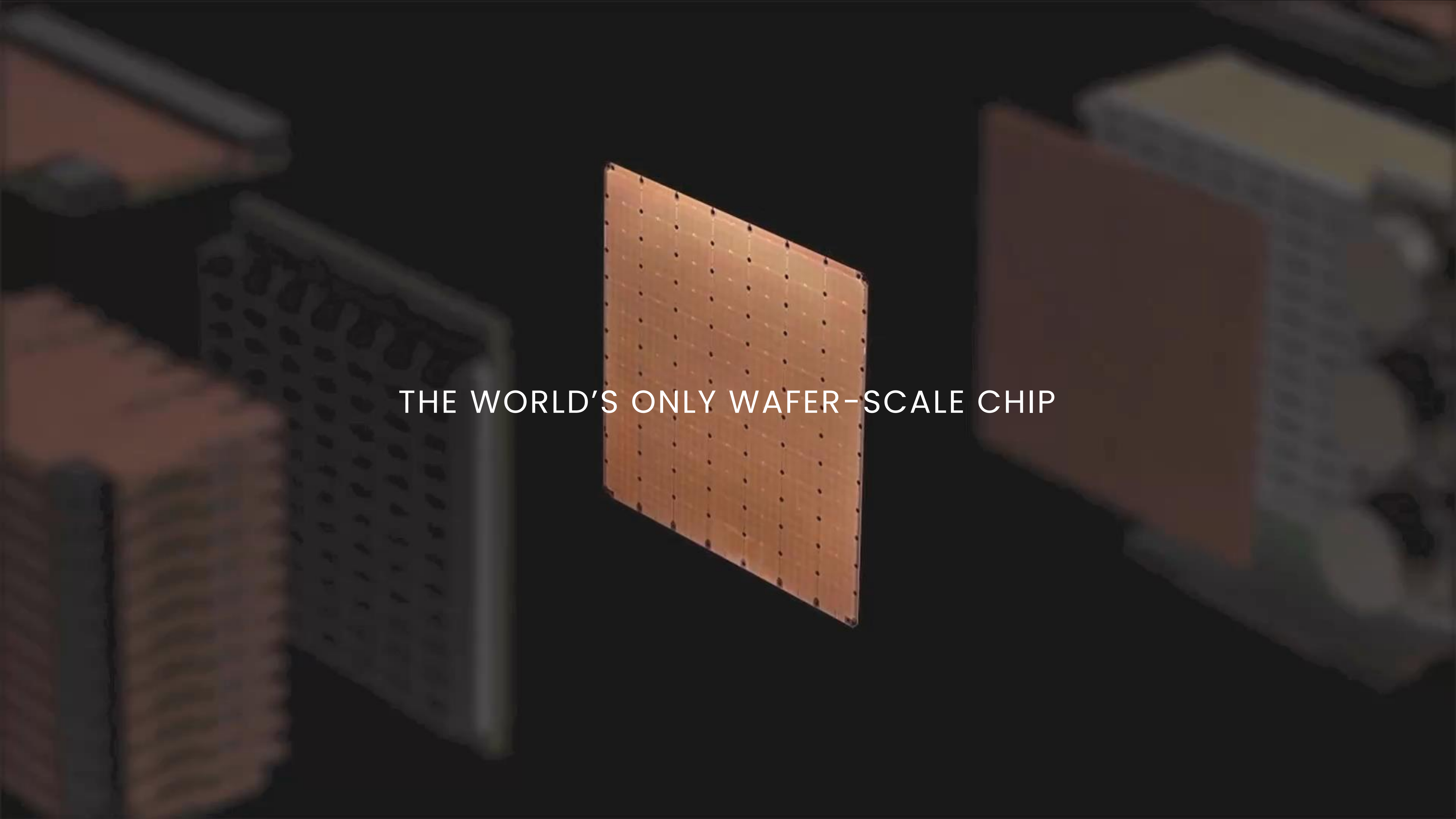
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THE WORLD'S ONLY WAFER-SCALE CHIP



INTRODUCING

THE CEREBRAS CS-4

THE CEREBRAS CS-4

UP TO

2x

FASTER TOKENS

UP TO

10x

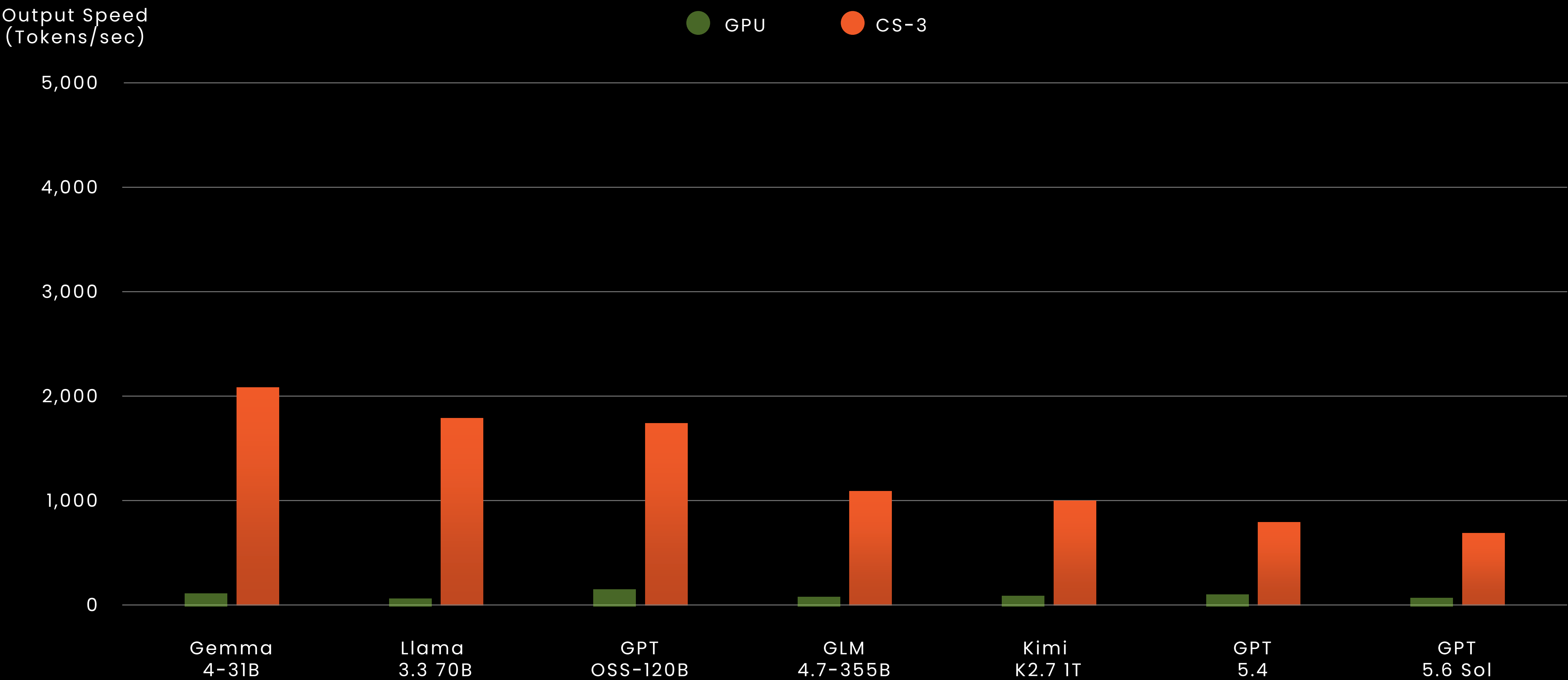
THROUGHPUT / WATT

VS. Industry-Leading CS-3

CS-4 Speed Advantage

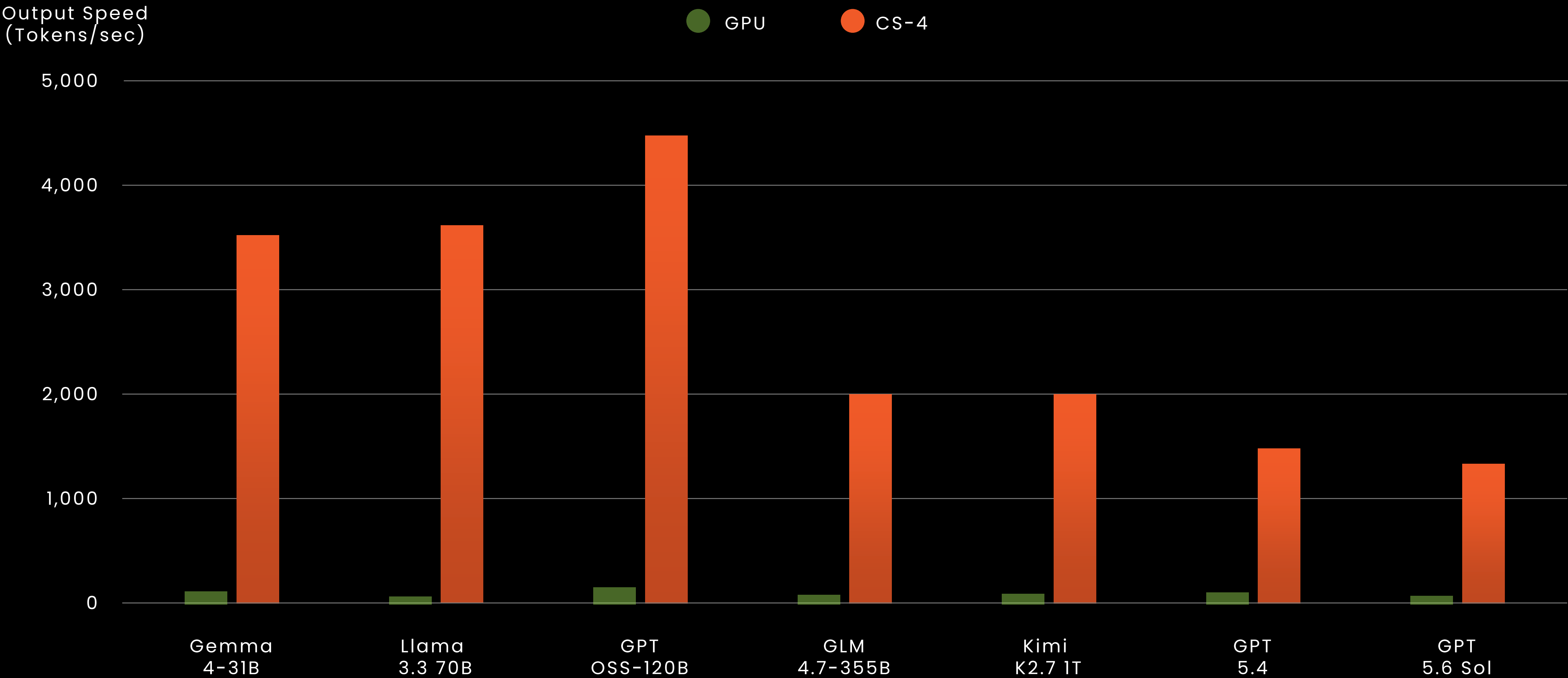


INFERENCE ON CEREBRAS IN PRODUCTION *TODAY*



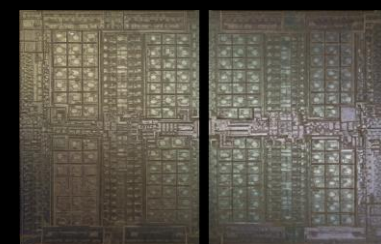
Source: Artificial Analysis and internal benchmarking (August 2026)

CS-4 IS TRANSFORMATIVE: *UP TO 30X FASTER*

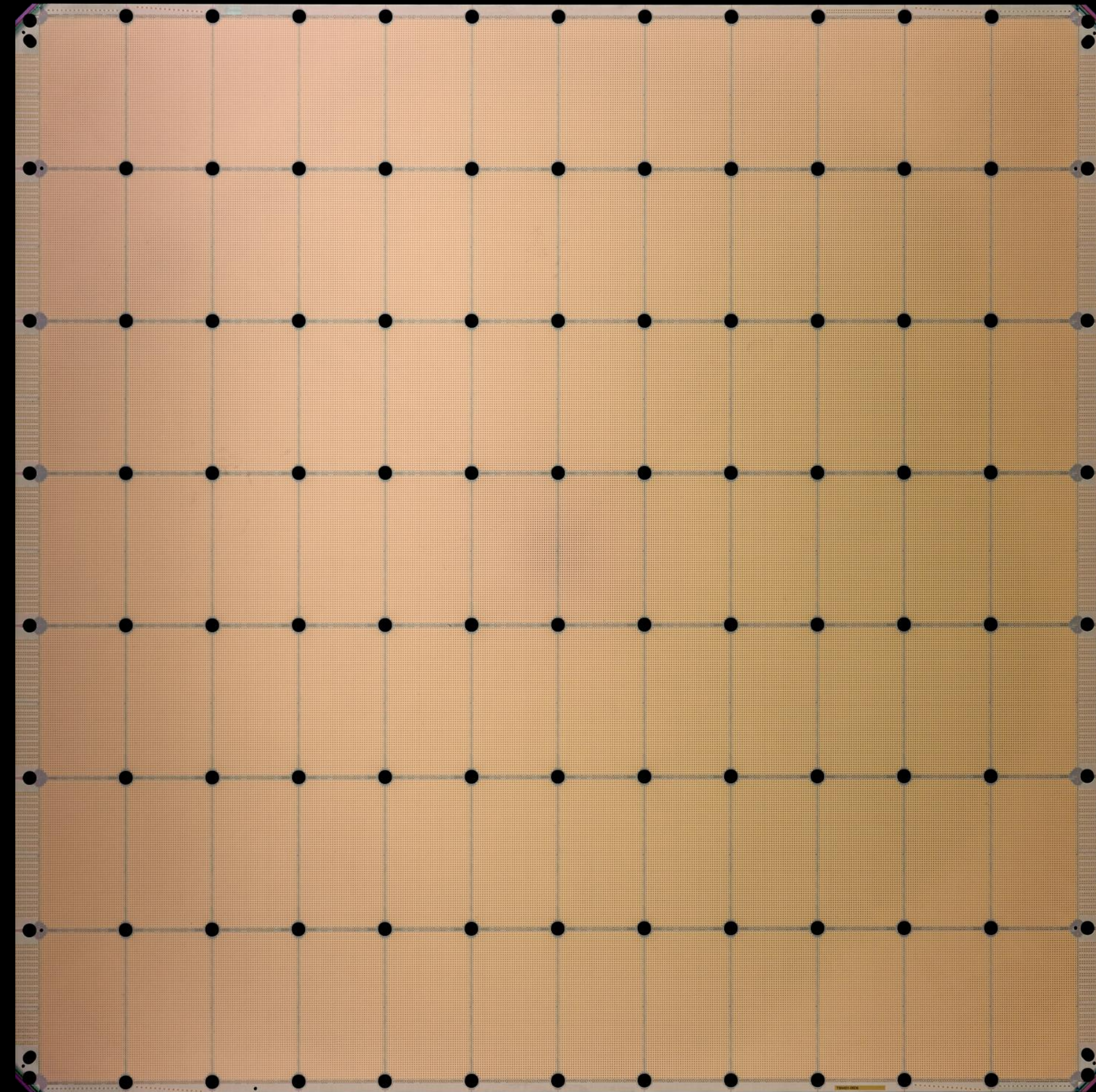


Source: Artificial Analysis and internal benchmarking (August 2026)

2,000x MORE MEMORY BANDWIDTH THAN RUBIN



NVIDIA Rubin
22 TByte/s



Cerebras CS-4 WSE-3T
43,200 TByte/s

NEXUS RACK-SCALE PLATFORM

DESIGNED FOR
MODULARITY

- 01 Simpler to build
- 02 Faster to deploy
- 03 Independent innovation
of power, compute, IO

POWER DELIVERY IN THE
FRONT OF THE RACK



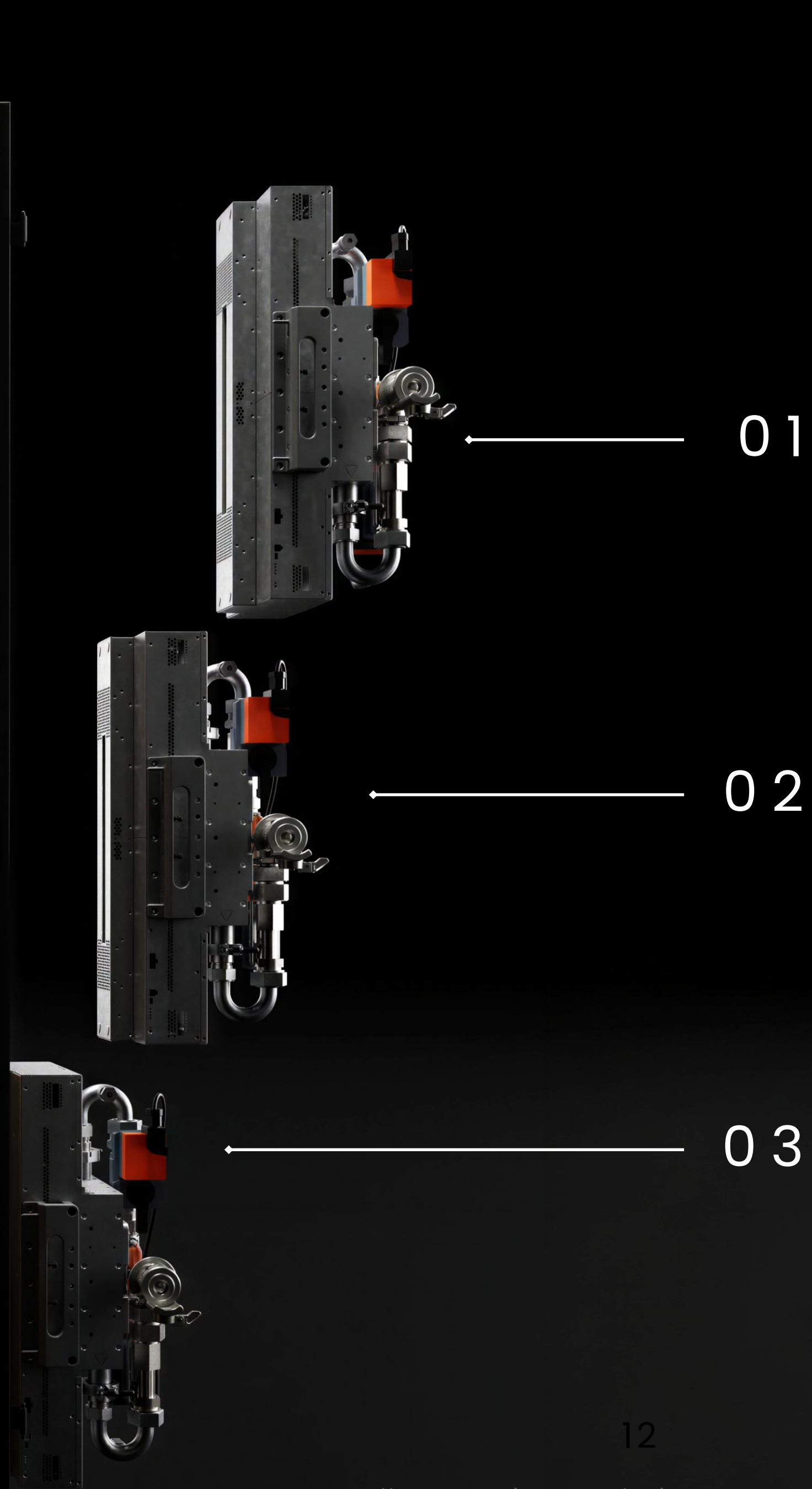
COMPUTE IN THE REAR IN
PLUGGABLE BACKPACKS

BACKPACK CONTAINS
WAFER SCALE ENGINE

All comparisons relative to Cerebras CS-3



3 WAFER-SCALE
BACKPACKS
PER CS-4



CS-4 BACKPACK:
NEXT-GEN PERFORMANCE

WAFER IO MODULE

2x more bandwidth
2x faster latency
Modular upgrades

WAFER PACKAGE

Direct vertical power
2x more power
2x more cooling

EFFICIENT MANUFACTURING

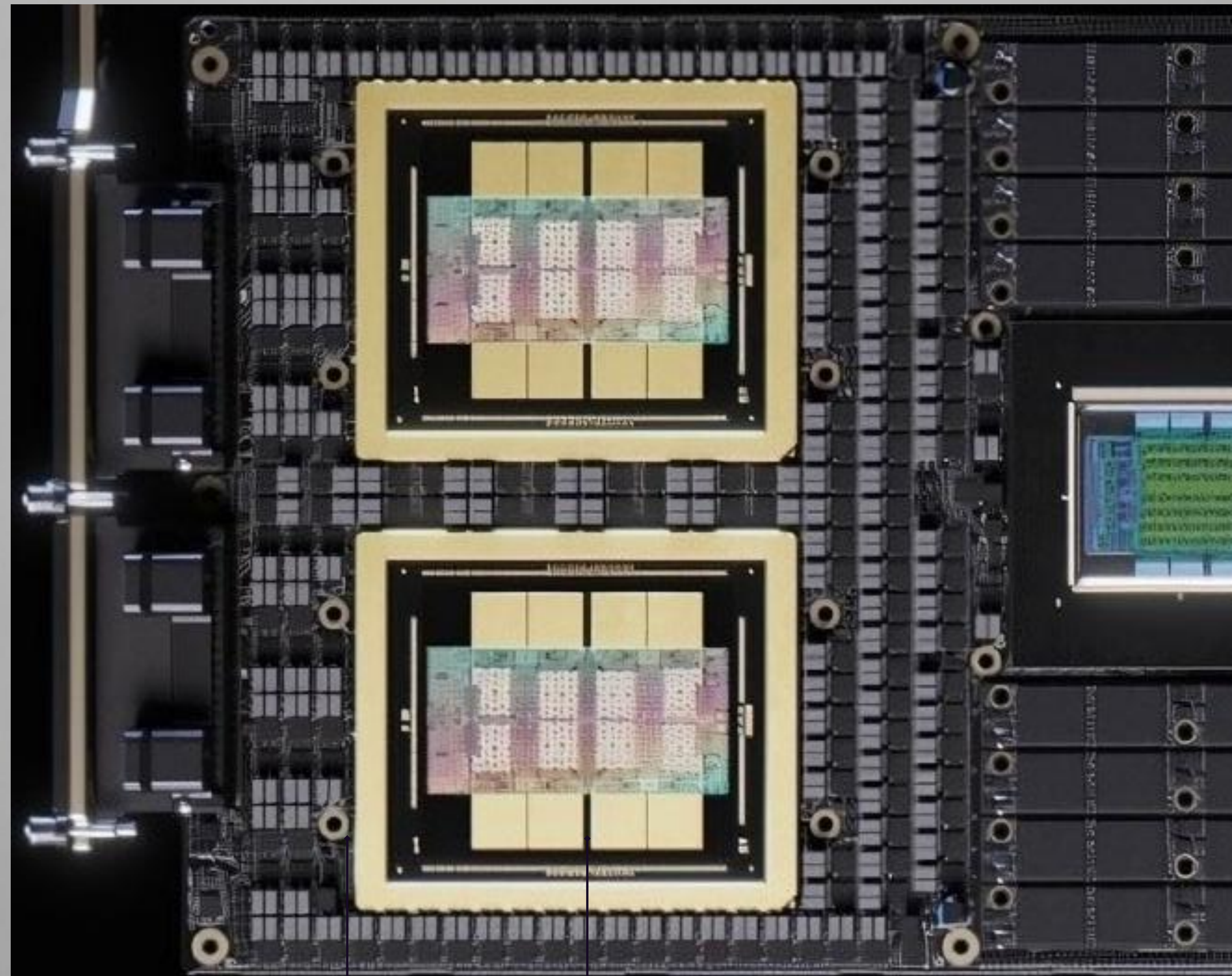
50% fewer components
60% more automation

INEFFICIENT POWER DISTRIBUTION OF GPU

~50MM BETWEEN POWER CONVERTERS AND SILICON LOAD

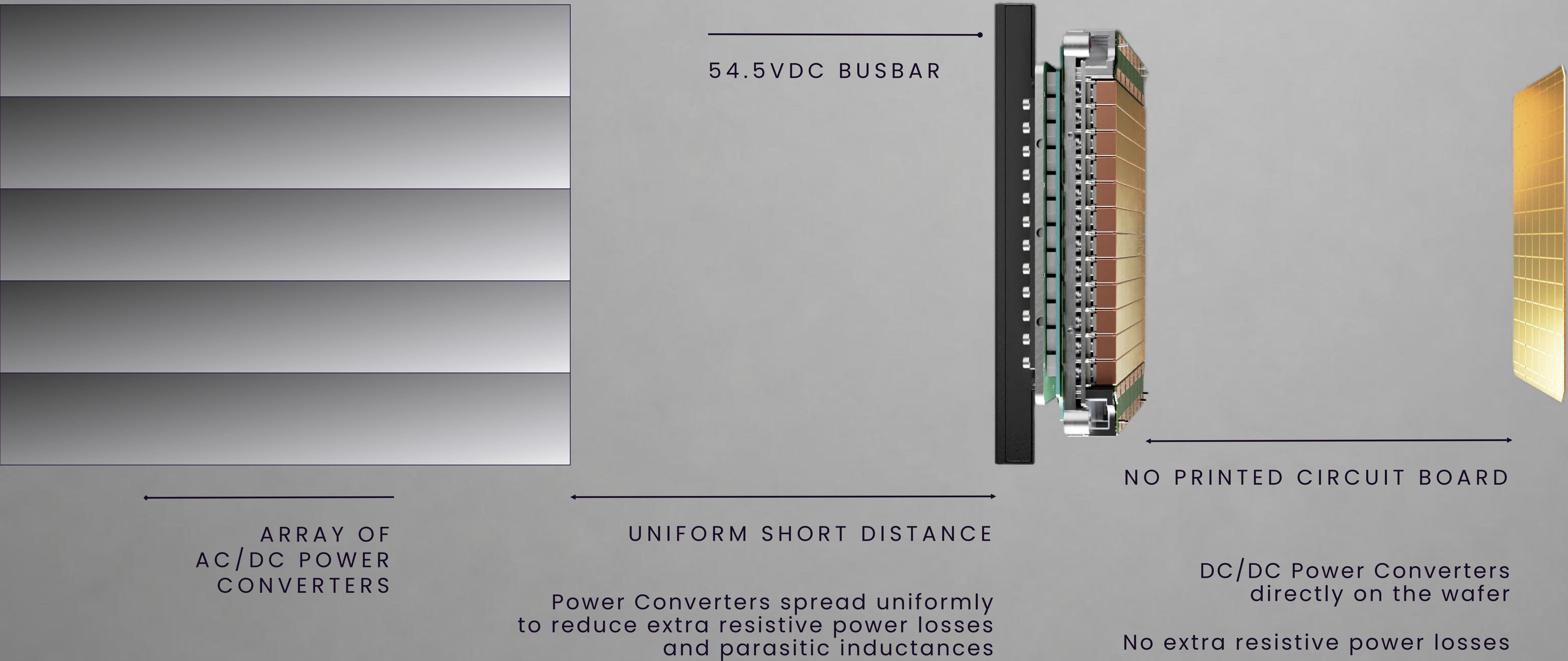
Power losses along path inside Printed Circuit Board

Requires more copper layers to reduce resistive losses — extra cost & complexity

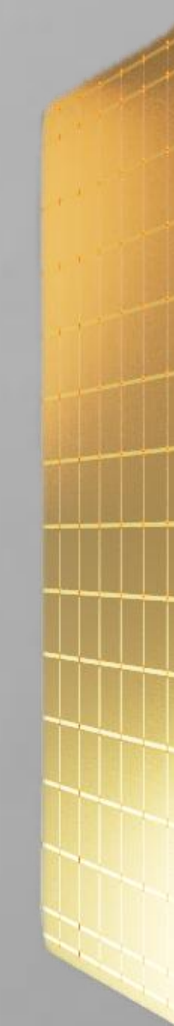
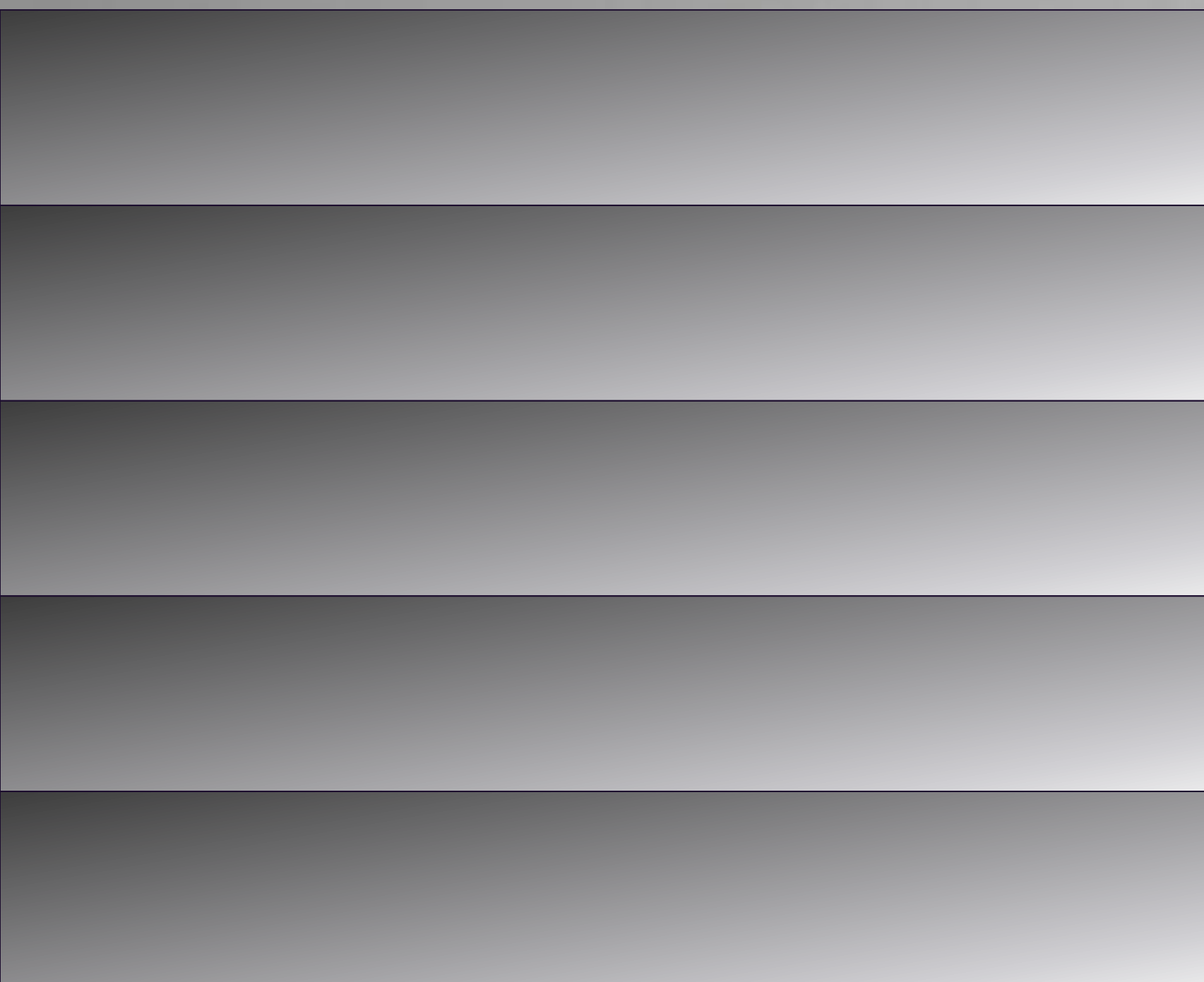


~50mm

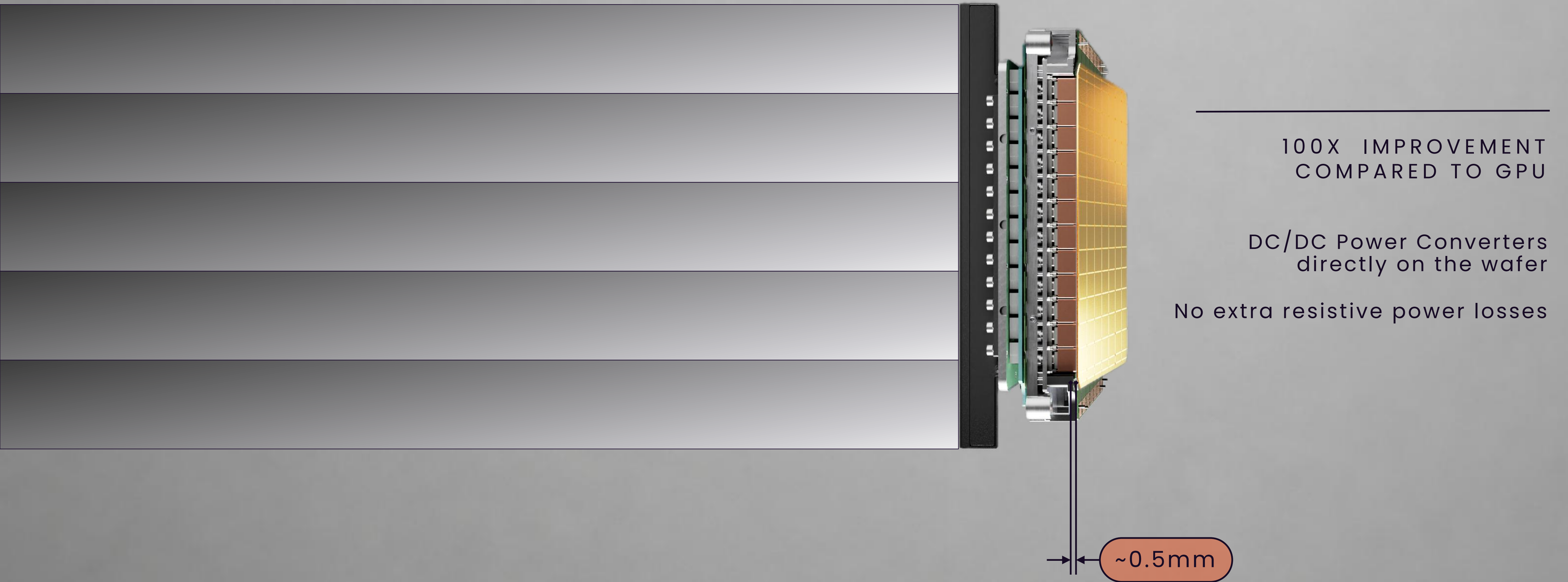
EFFICIENT POWER DISTRIBUTION



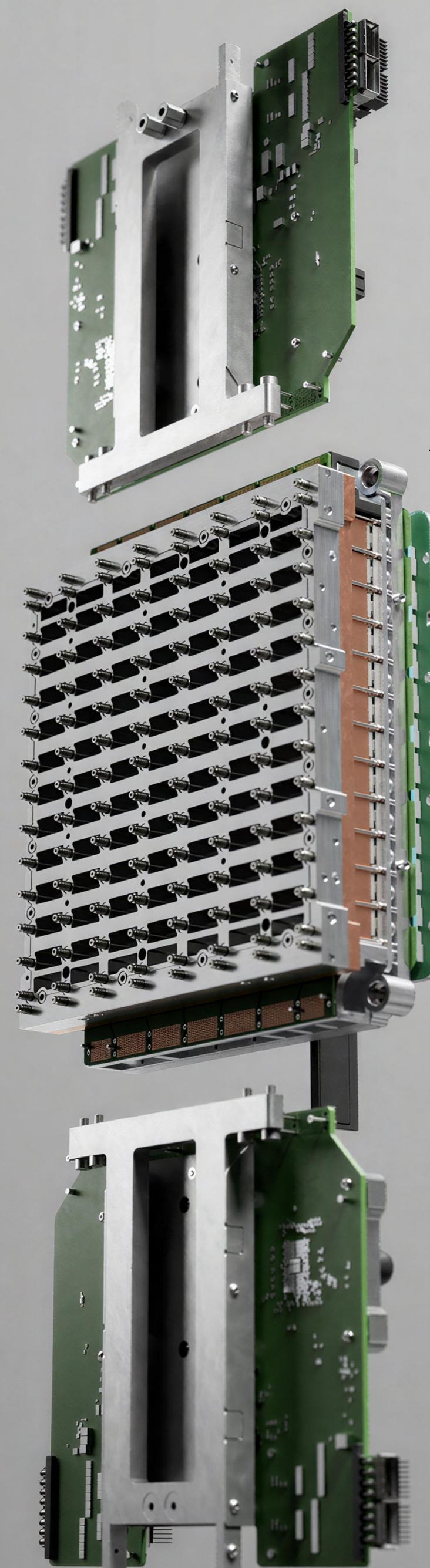
EFFICIENT POWER DISTRIBUTION



EFFICIENT POWER DISTRIBUTION



THE NEXT GEN WAFER IO INTERFACE



HIGHER BANDWIDTH
AND LOWER LATENCY

New direct wafer link interfaces
Standard RoCE protocol network

NEW WAFER IO MODULE

Extends fabric from wafer edges
Modular & programmable for the future

INTEGRATED WATER CONDITIONING

FLOW REGULATION ACTUATOR

Guarantees proper water flow rate

LEAK DETECTION MODULE

Monitors leaks and condensation

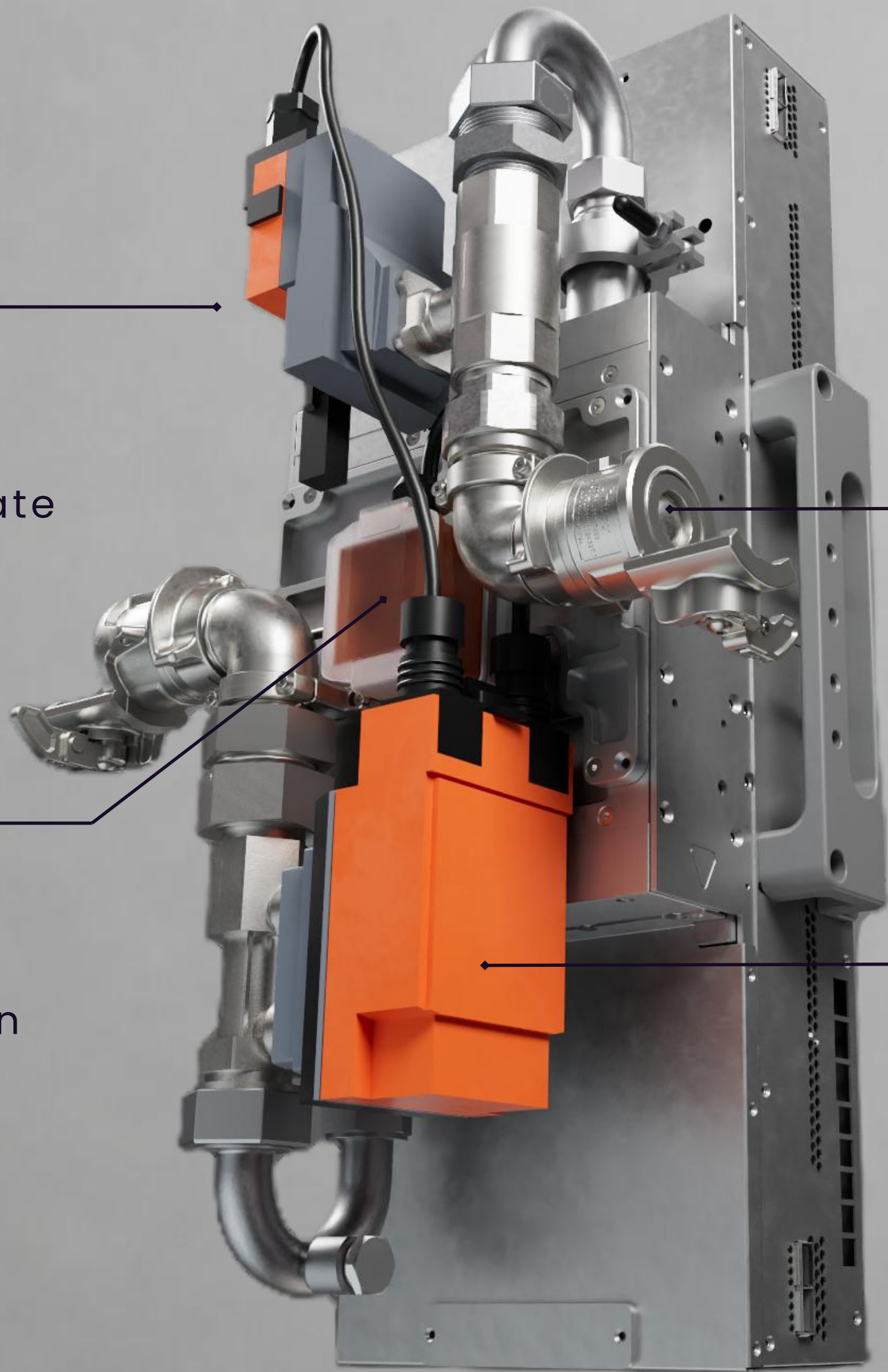
Shuts off AC power to all the
Power Supply modules

VALVED DRY QUICK DISCONNECTS

For easy install

ENERGY METER

Monitors water flow rate
Monitors water inlet & outlet temp.



WATER & COMPUTE IN THE REAR

INTEGRATED WATER DELIVERY

Supply and return manifolds on each side
Valved Dry Quick Disconnects for easy install
isolated from high-voltage AC power section in the front

CS-4 BACKPACK

Backpack can easily be swapped

INTEGRATED FIBER CONDUIT

Fiber optic cables are protected



POWER IN THE FRONT

AC/DC POWER SUPPLY MODULES

Input: up to 277VAC, Output 54.5VDC

Up to 30x modules per backpack

Flexible power feed redundancy:
5+1, 4+1, 3+1, 4+2

Air-cooled

FAN MODULES

Cool small devices in the Backpack



INTEGRATED POWER PROTECTION

Each Power Supply is
protected by a 30A circuit
breaker behind cover panels

POWER IN THE FRONT

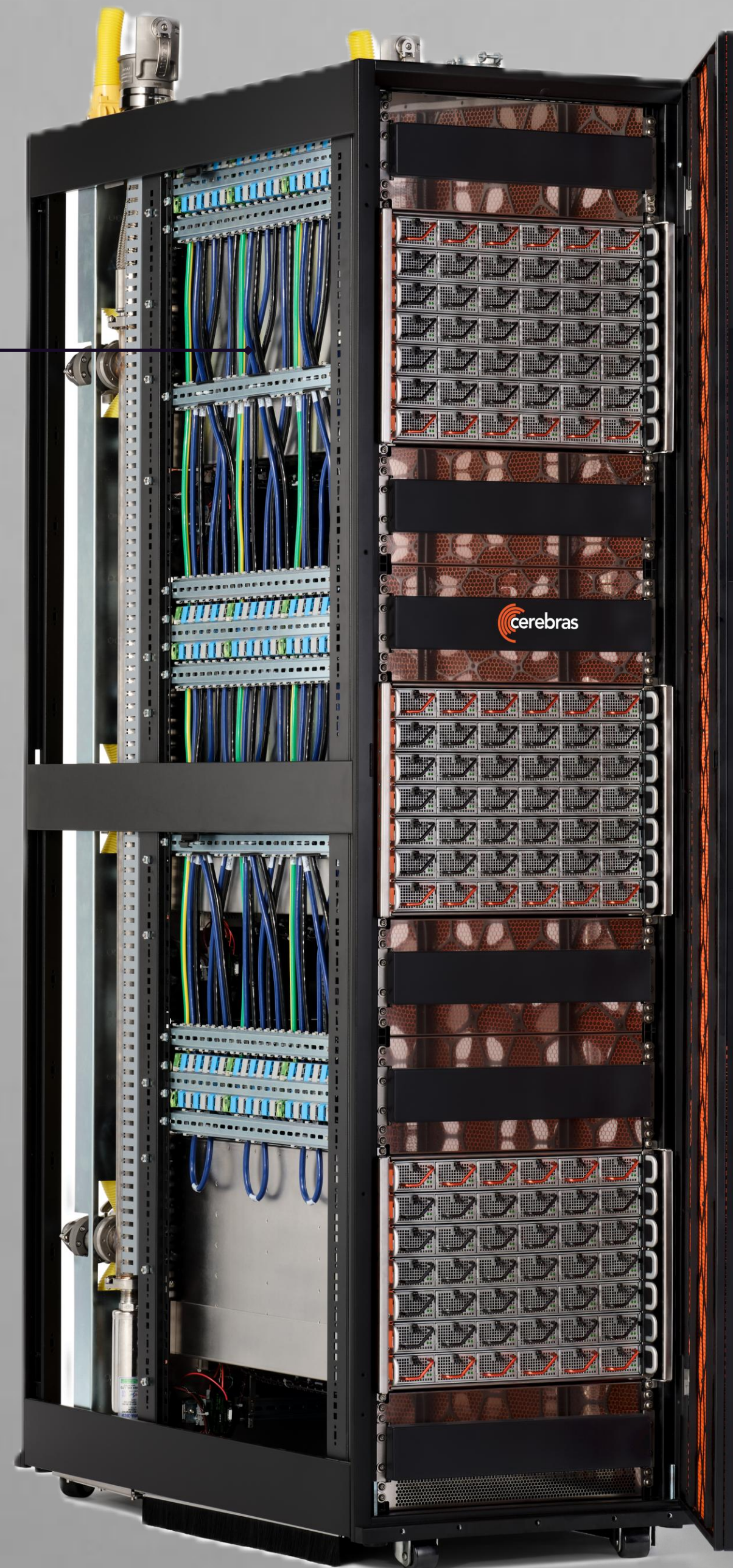


POWER IN THE FRONT

INTEGRATED POWER DELIVERY

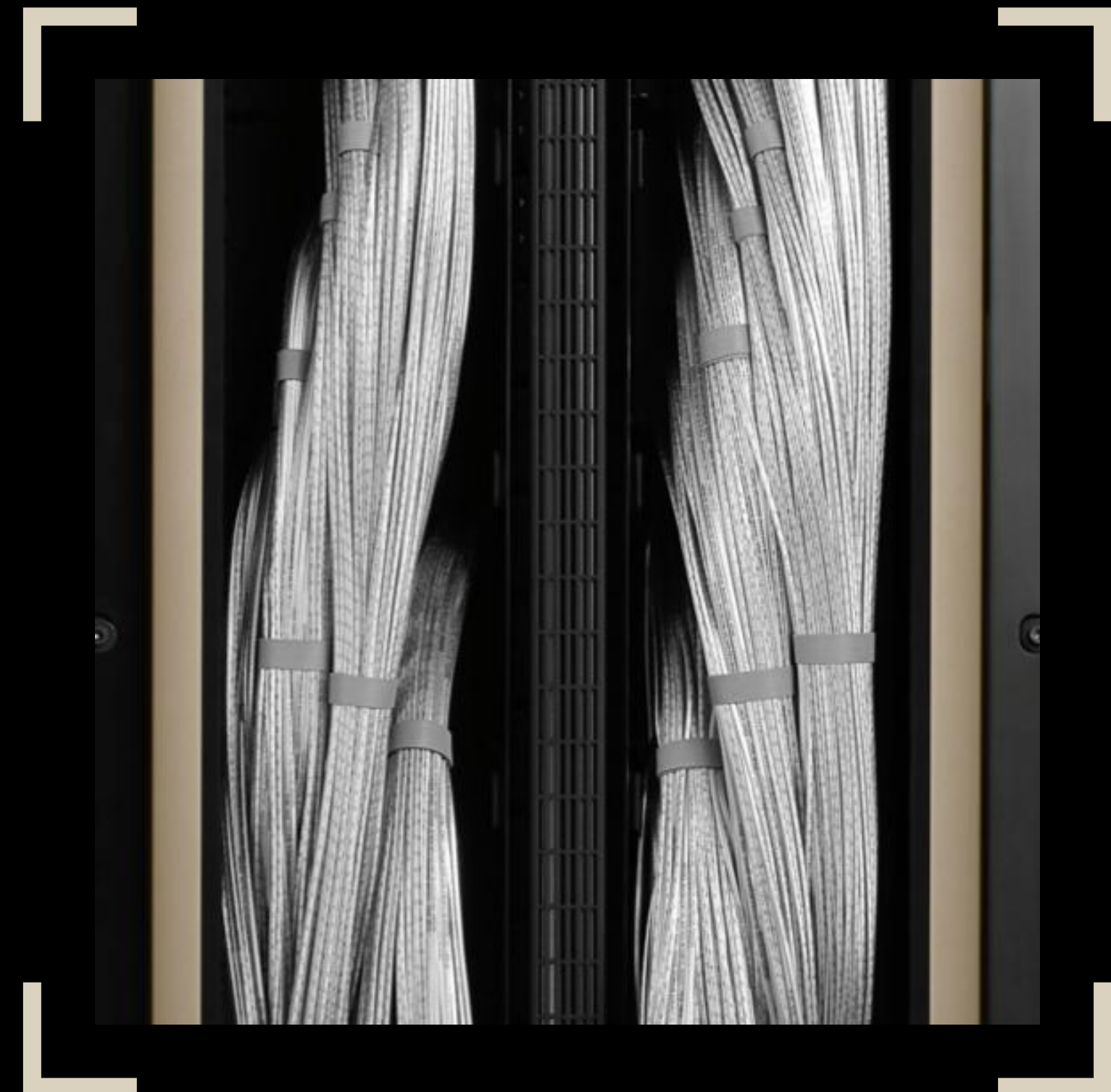
Power is fed by up to 6 AC feeds from the top of the rack

Fully phase balanced AC power delivery interconnect inside both sides of the rack

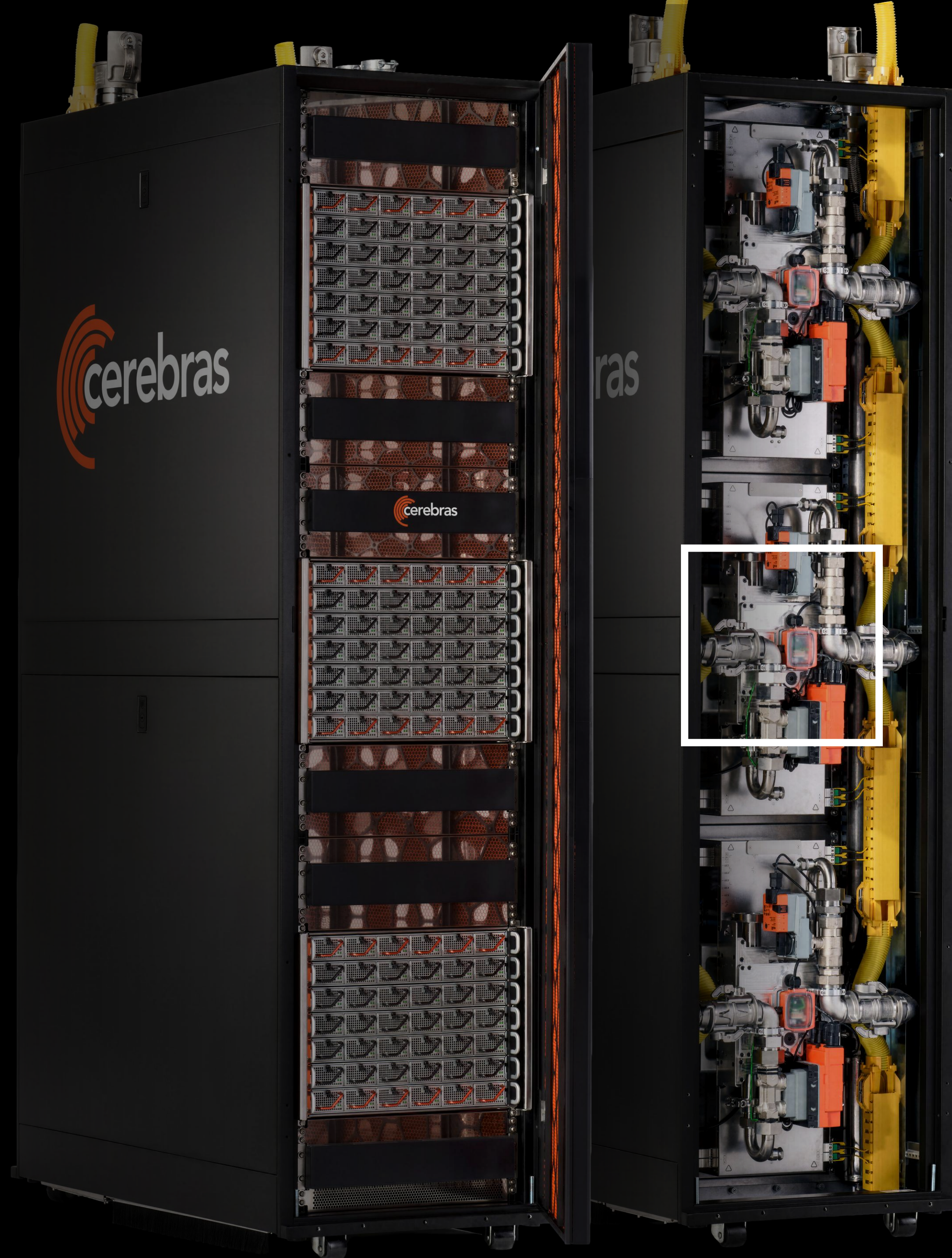




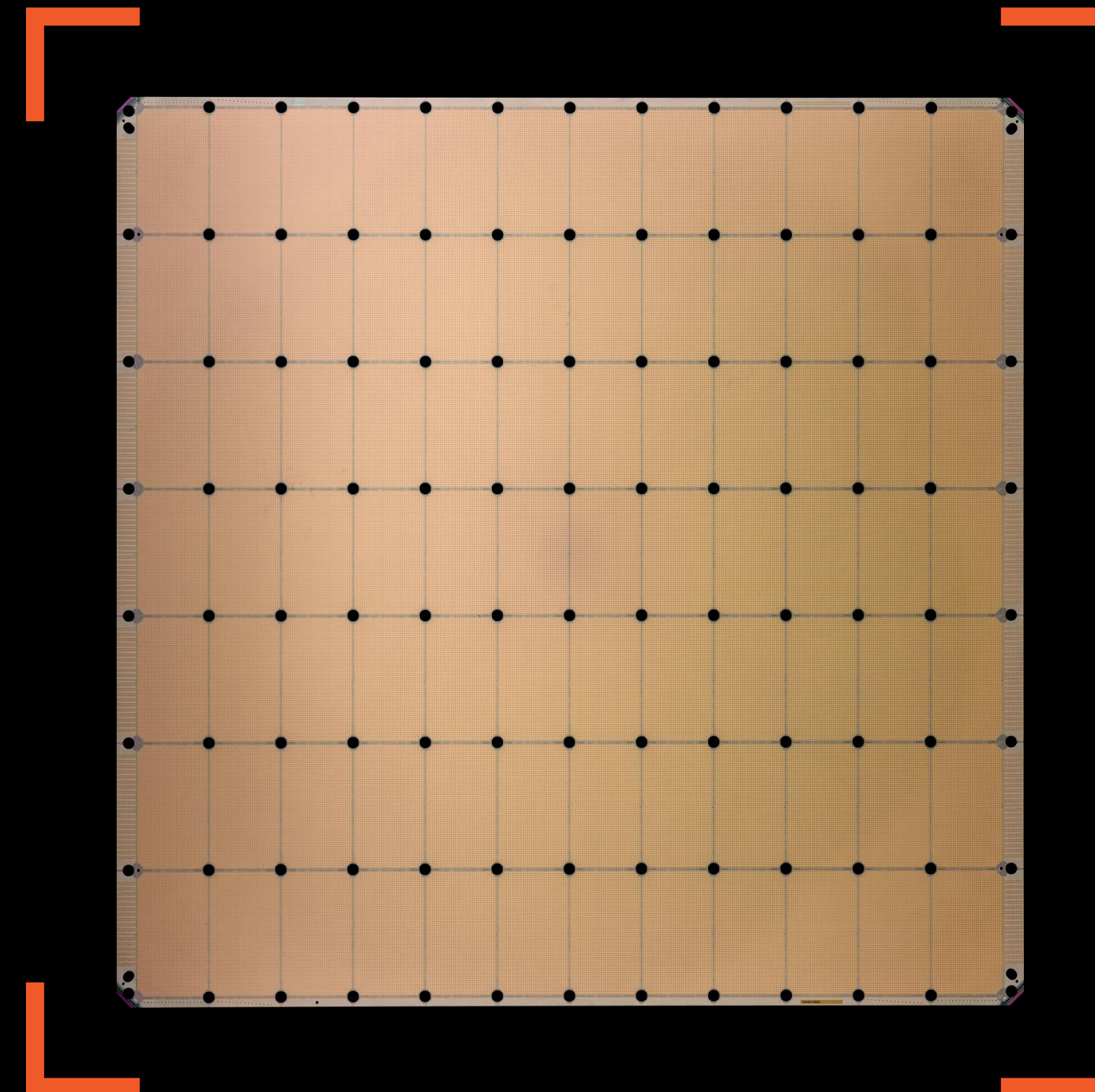
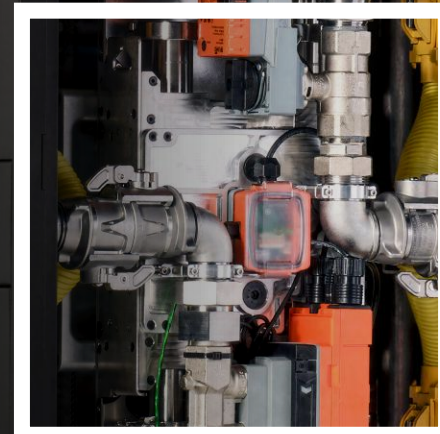
RUBIN NVL72
260 TB/S NVLINK
5,000 CABLES



“MORE BANDWIDTH
THAN THE INTERNET”



CS-4 WSE-3T
53.5 PB/S WAFER FABRIC
NO CABLES, ALL ON-CHIP



200X HIGHER BANDWIDTH THAN
GPU INTERCONNECT

HIGHER BANDWIDTH
LOWER LATENCY
WITHIN EACH WAFER

2X

BANDWIDTH PER WAFER

- ✦ New higher speed wafer links
- ✦ 2.4Tb/s aggregate bandwidth

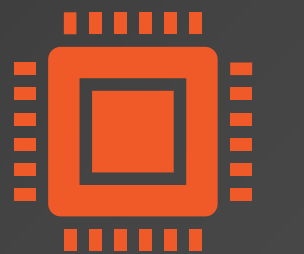
3us latency

CS-4 CLUSTER

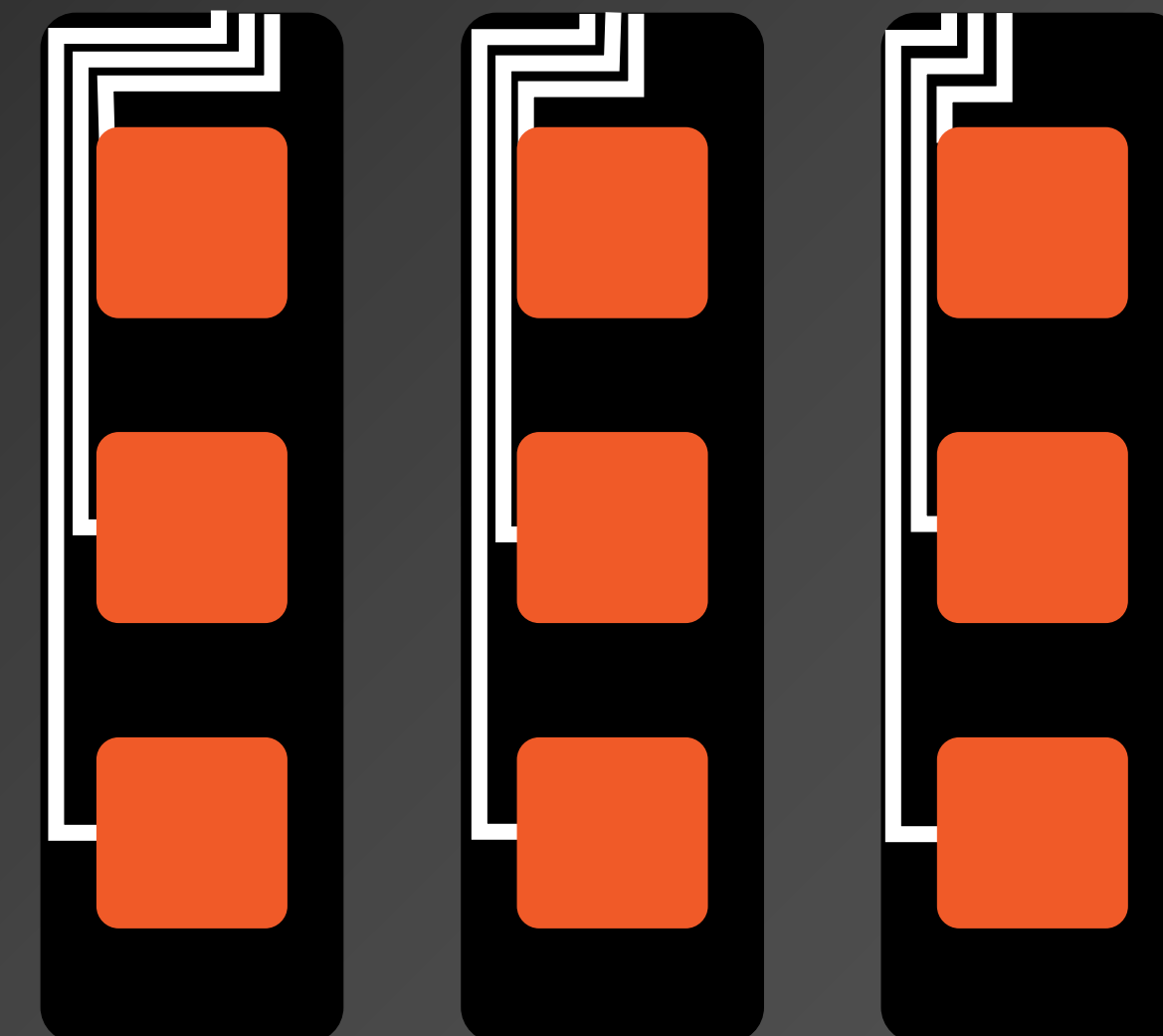
NETWORK



USERS



GPU / ASIC



HIGHER BANDWIDTH
LOWER LATENCY
BETWEEN WAFERS

2X

BANDWIDTH PER WAFER

- ✦ New higher speed wafer links
- ✦ 2.4Tb/s aggregate bandwidth

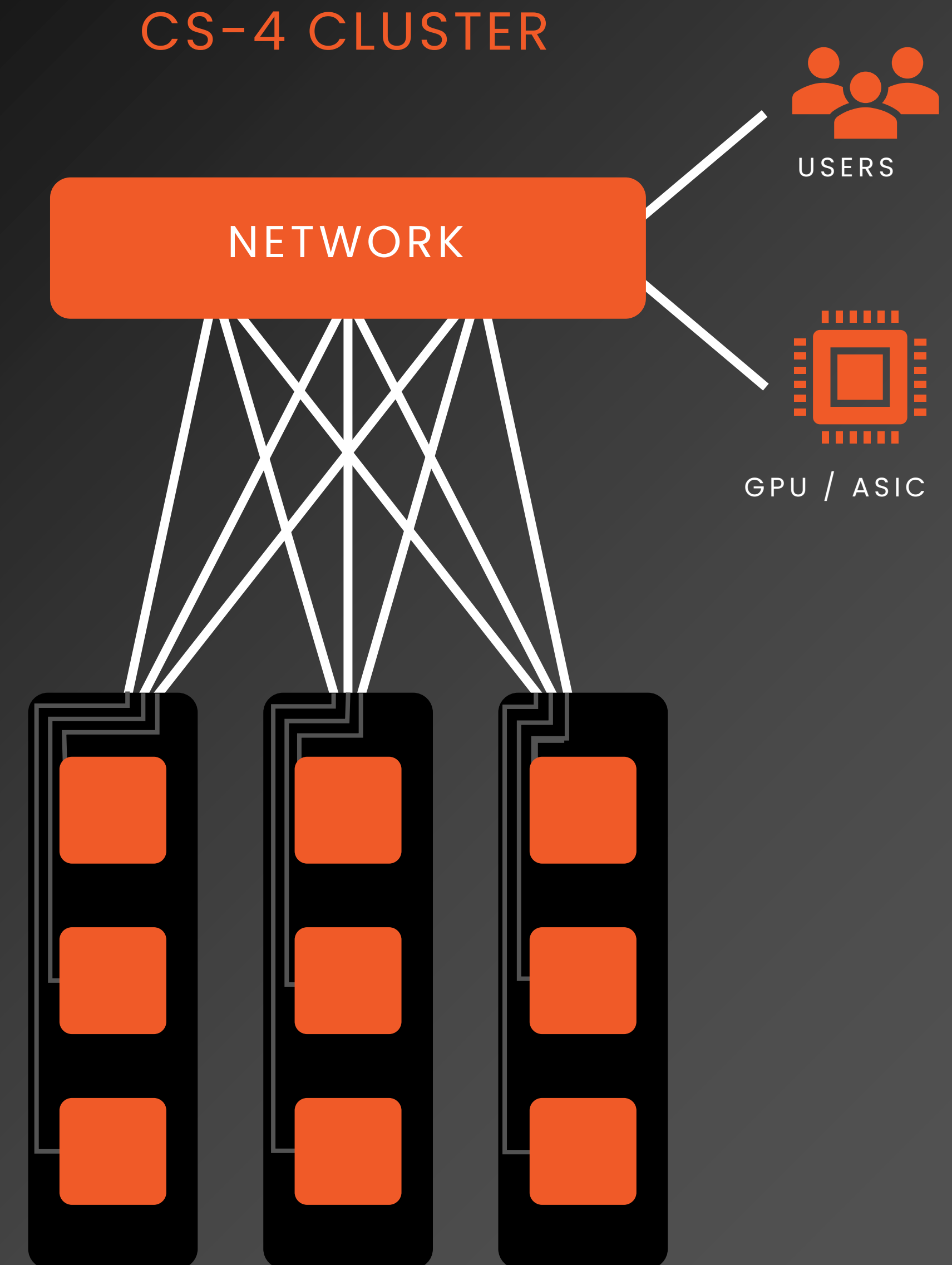
1.7X

FASTER NETWORK LATENCY

- ✦ New low latency packet pipeline
- ✦ 3us latency to users & disaggregated devices

3us latency

2.4 Tb/s per wafer



HIGHER BANDWIDTH
LOWER LATENCY
BETWEEN WAFERS

2X

BANDWIDTH PER WAFER

- ✦ New higher speed wafer links
- ✦ 2.4Tb/s aggregate bandwidth

1.7X

FASTER NETWORK LATENCY

- ✦ New low latency packet pipeline
- ✦ 3us latency through network

2.5X

FASTER WAFER-TO-WAFER

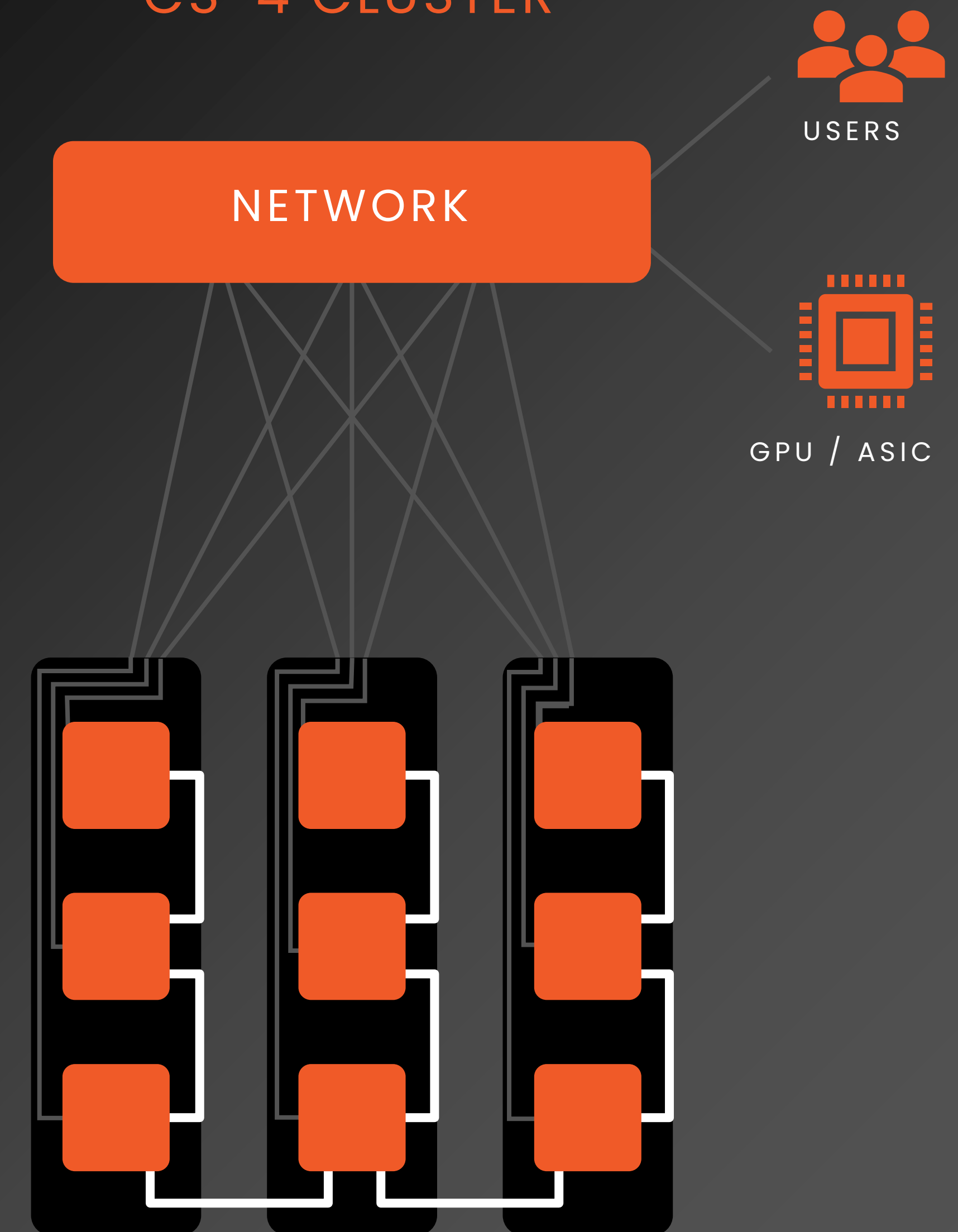
- ✦ New Direct Wafer Links
- ✦ 2us latency wafer-to-wafer

3us latency

2.4 Tb/s per wafer

2us latency

CS-4 CLUSTER

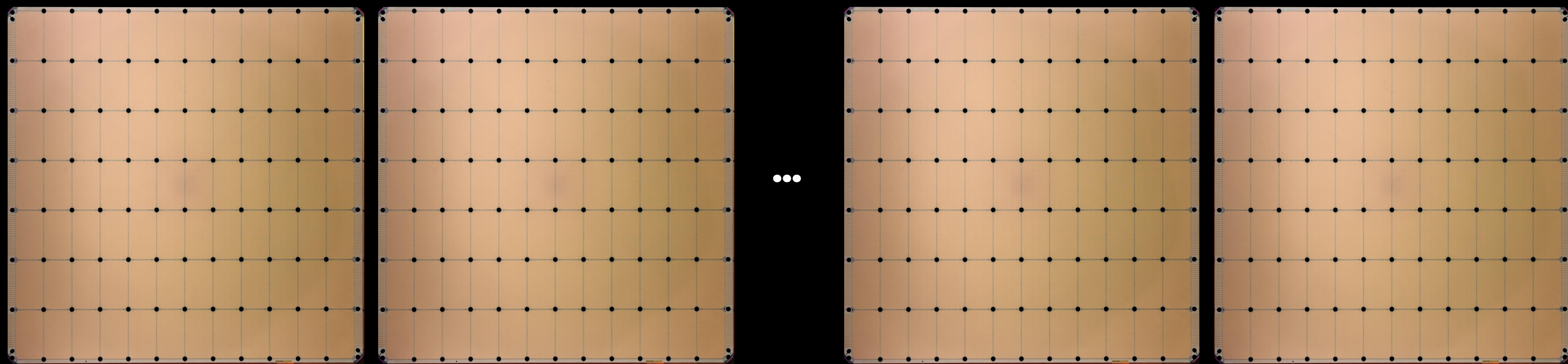
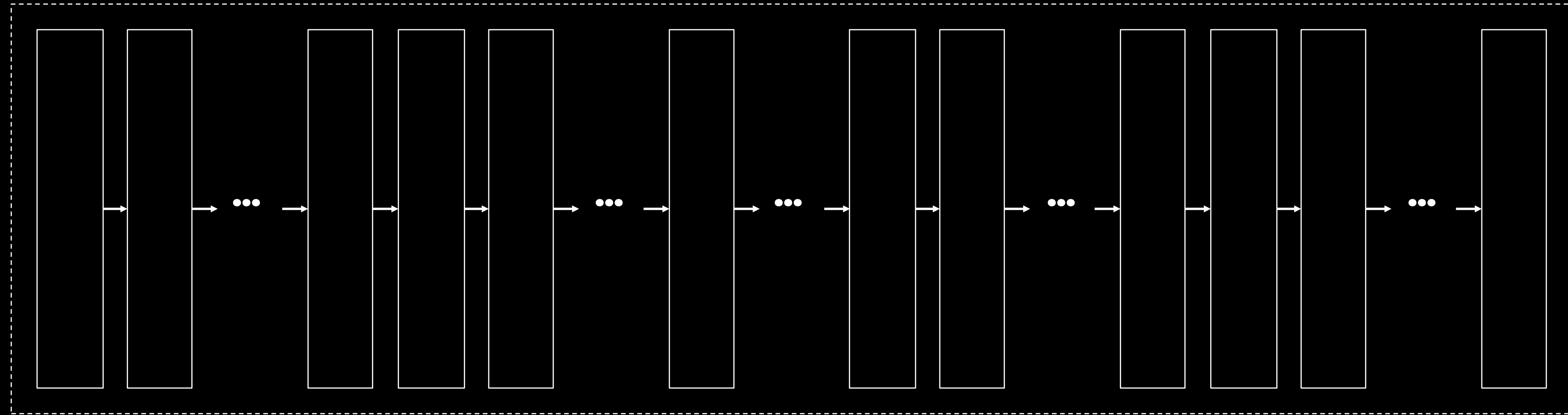


CS-4 BY THE NUMBERS

	CS-3	CS-4
AI Compute	125 PFLOPS	750 PFLOPS
Wafers	1x WSE-3	3x WSE-3 Turbo
Memory capacity	44 GByte	132 GByte
Memory bandwidth	21.6 PByte/s	129.6 PByte/s
Fabric bandwidth	26.7 PByte/s	160.5 PByte/s
IO bandwidth	1.2 Tbit/s	7.2 Tbit/s
IO latency	5 us	2 us

CS-3 ALREADY RUNS THE LARGEST FRONTIER MODELS ULTRAFAST

GPT-5.6 SOL



CS-3 CLUSTER

CS-3 ALREADY RUNS THE LARGEST FRONTIER MODELS ULTRAFAST

GPT-5.6 SOL

01

Pipelining large models
across wafers is seamless
and fast

02

Keep almost all high communication
on wafer, using internal high
bandwidth fabric

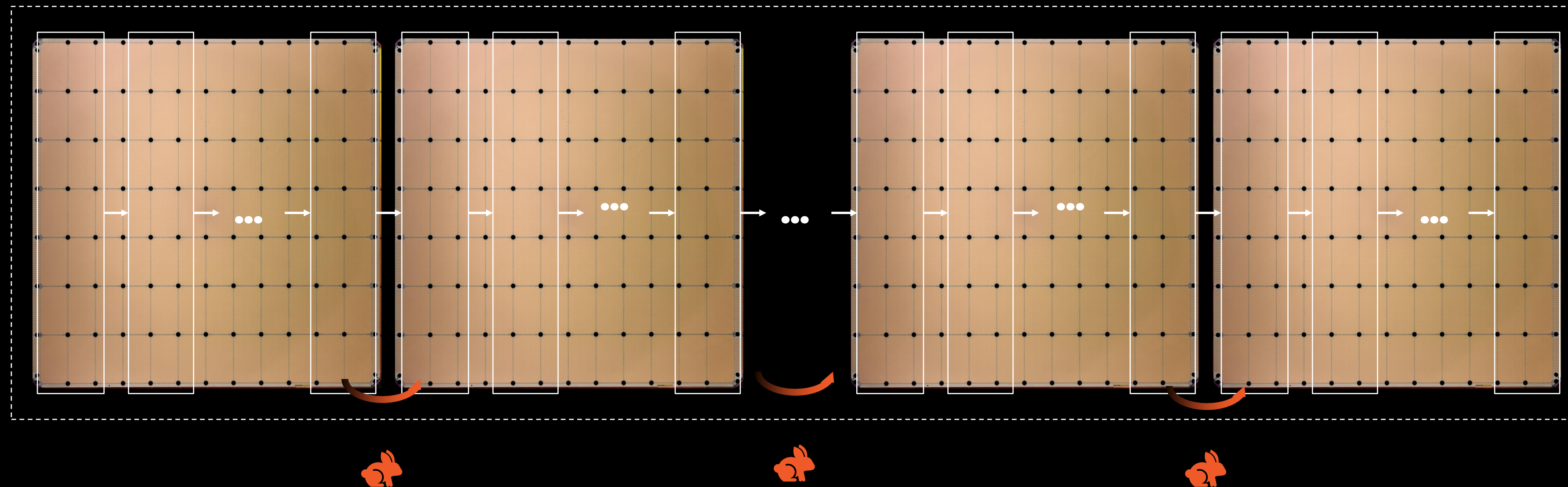
03

Transfer only activations
between wafers, requiring
relatively lower bandwidth

GPT-5.6 SOL

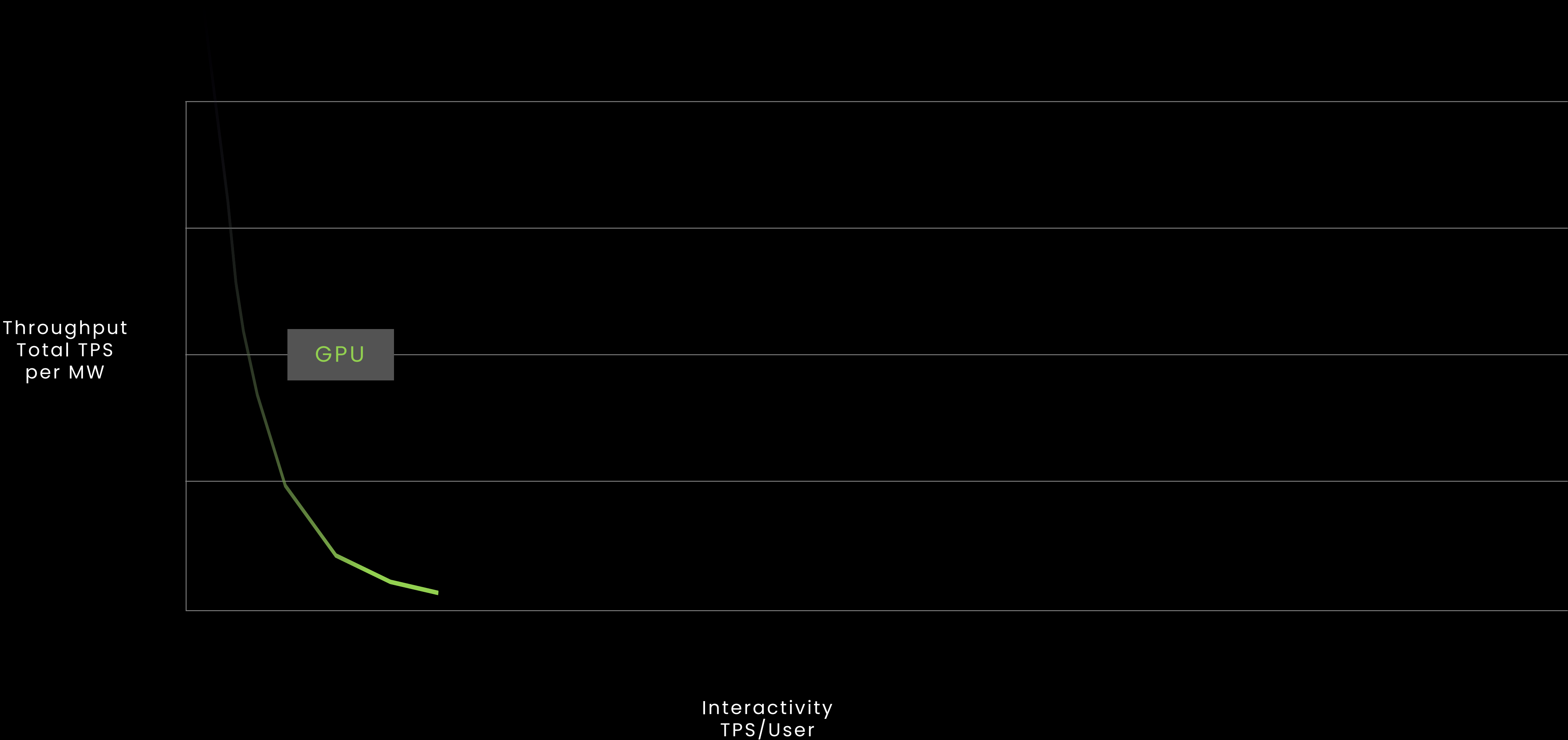
X

CS-3 CLUSTER



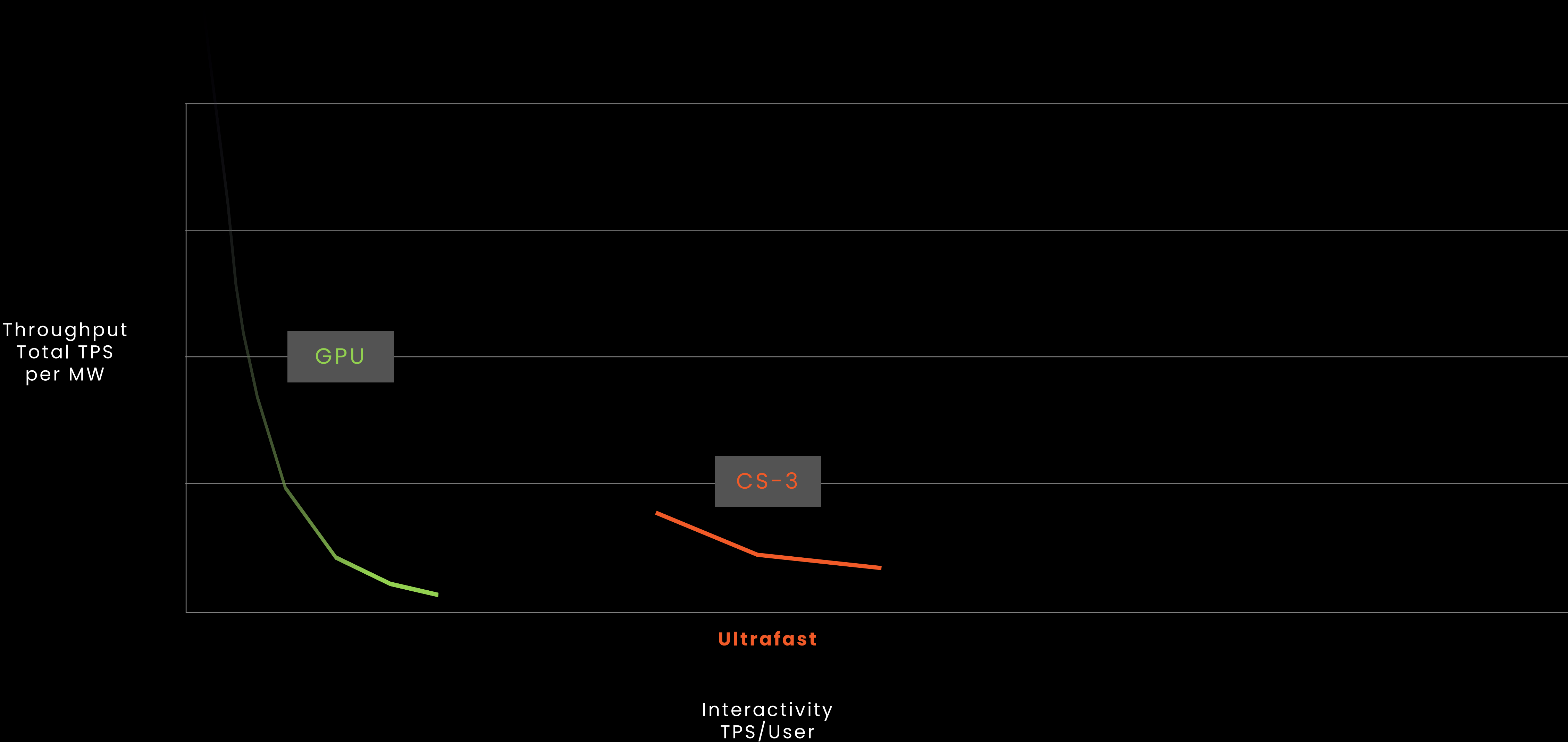
Higher I/O bandwidth means faster hops between wafers for **< 0.2 ms latency for 10T parameter model**

GPU IS SLOW AT INFERENCE



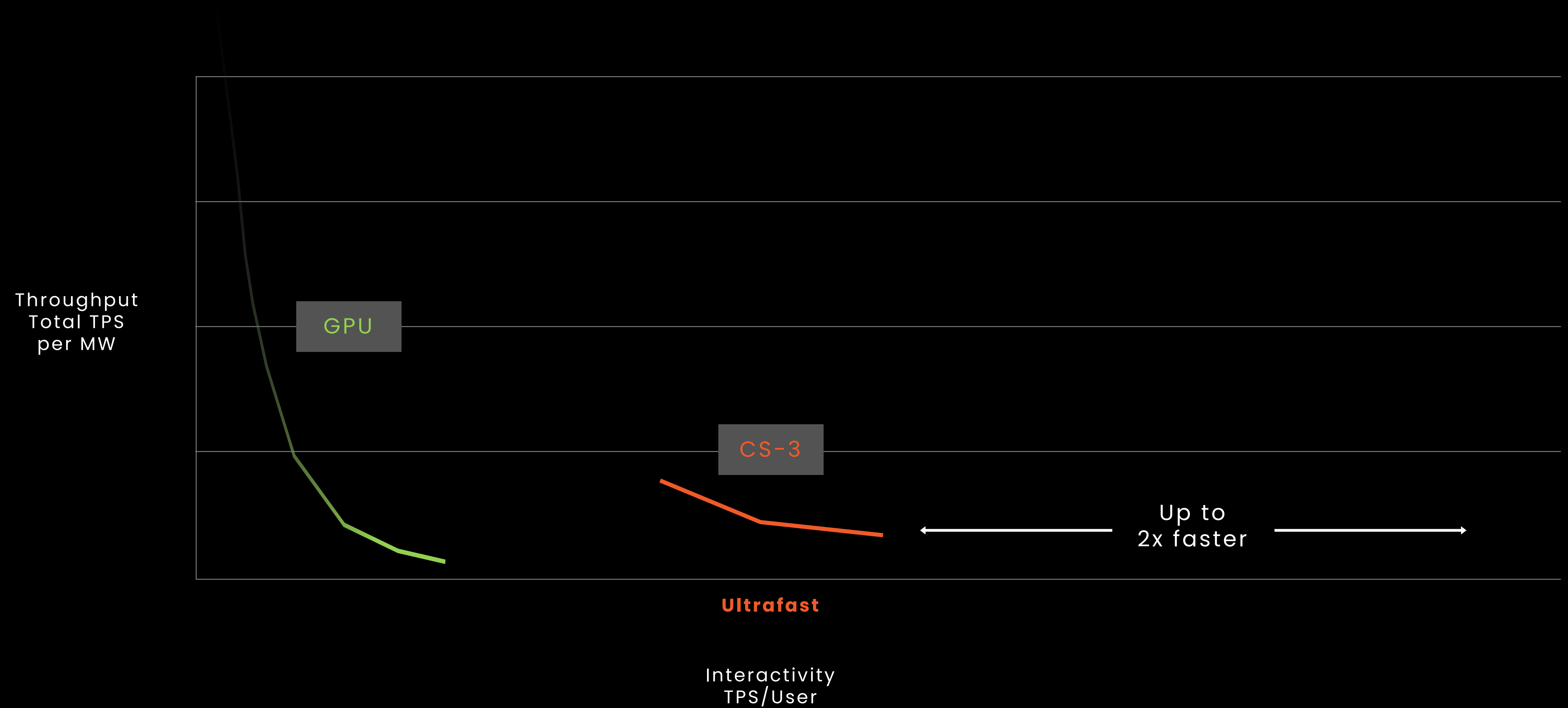
Source: Artificial Analysis and internal benchmarking (August 2026)

STARTING FROM UP TO 15x FASTER THAN GPUS



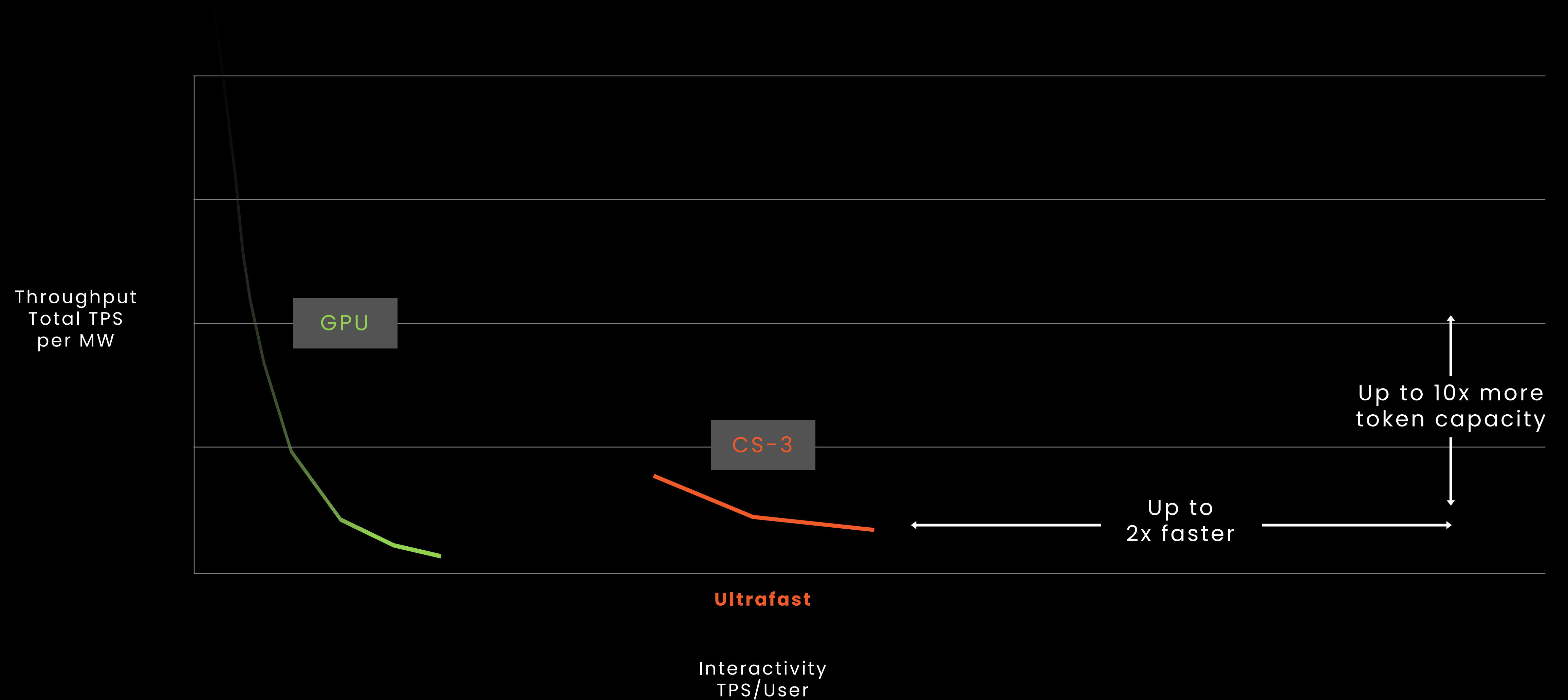
Source: Artificial Analysis and internal benchmarking (August 2026)

THE FASTEST GETS FASTER



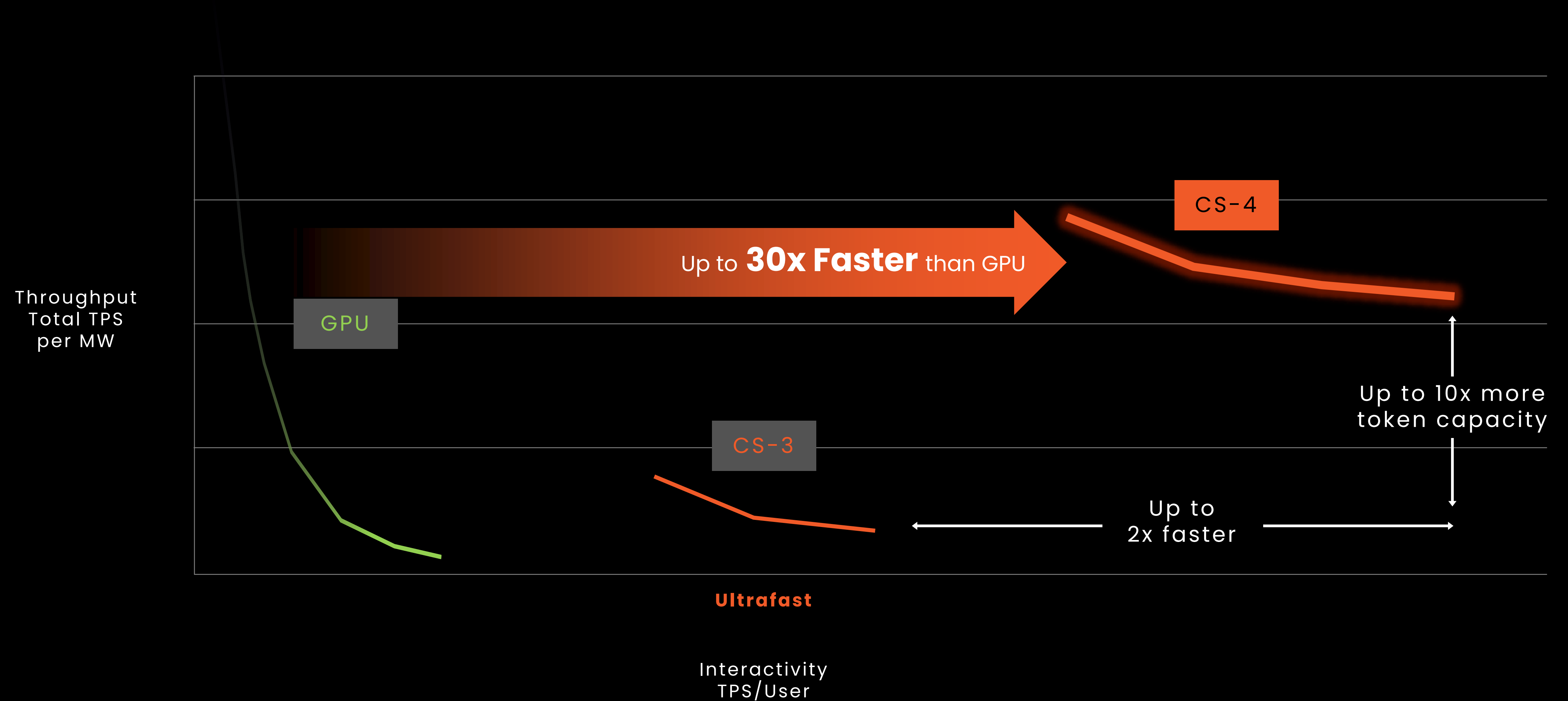
Source: Artificial Analysis and internal benchmarking (August 2026)

THE FASTEST GETS FASTER. BUILT FOR HYPERSCALE.



Source: Artificial Analysis and internal benchmarking (August 2026)

THE FASTEST GETS FASTER. BUILT FOR HYPERSCALE.

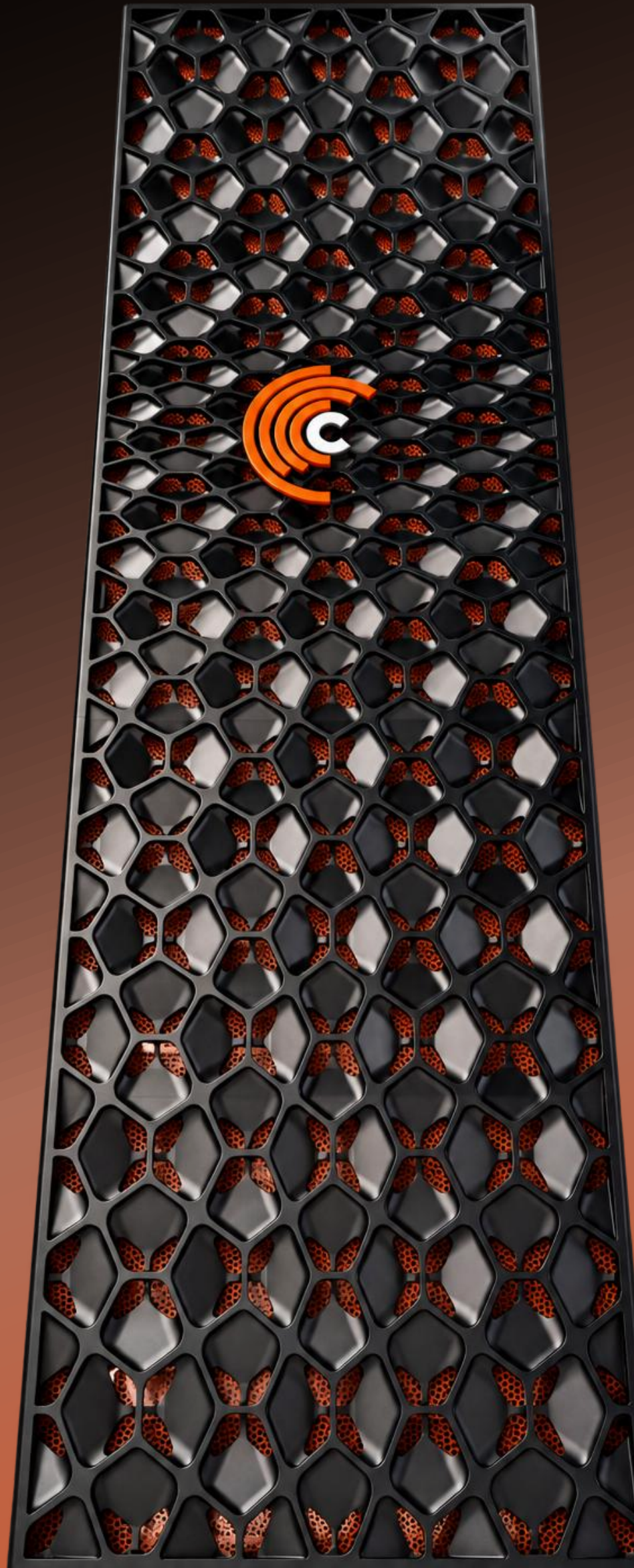


Source: Artificial Analysis and internal benchmarking (August 2026)

NEXUS RACK-SCALE PLATFORM

CS-4

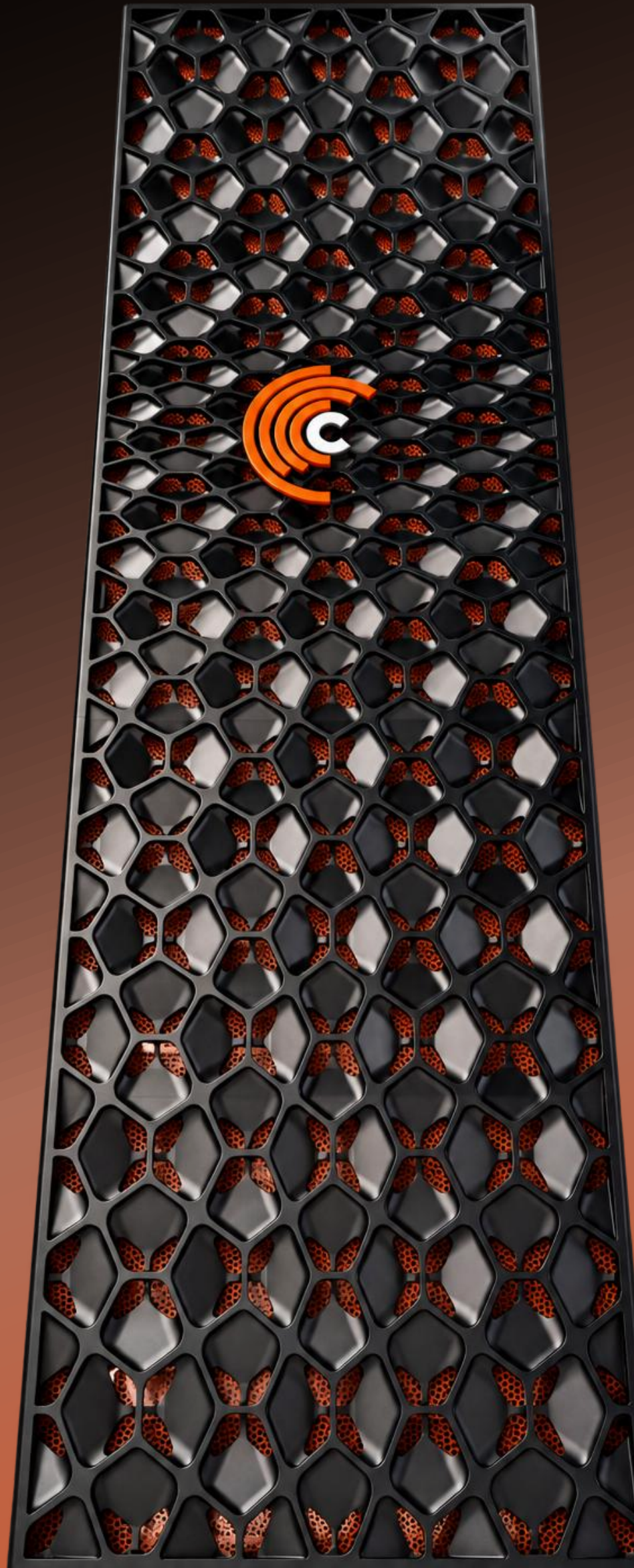
**EARLY ACCESS NOW,
GA LATER IN Q3 2026**



EXTENDING CEREBRAS PERFORMANCE LEADERSHIP

CS-4

**EARLY ACCESS NOW,
GA LATER IN Q3 2026**



CS-5 • CS-6

**DESIGNED FOR
MULTIPLE
GENERATIONS**

CEREBRAS CS-5 IN 2027

DESIGNED TO SET ANOTHER STANDARD OF SPEED & EFFICIENCY

From 30B models to the largest multi-trillion parameter models

Gemma 4 31B, gpt-oss 120B, ...

DeepSeek, Kimi, GPT 5.6 Sol, ...

UP TO

10,000

TPS / USER

UP TO

5,000

TPS / USER

UP TO

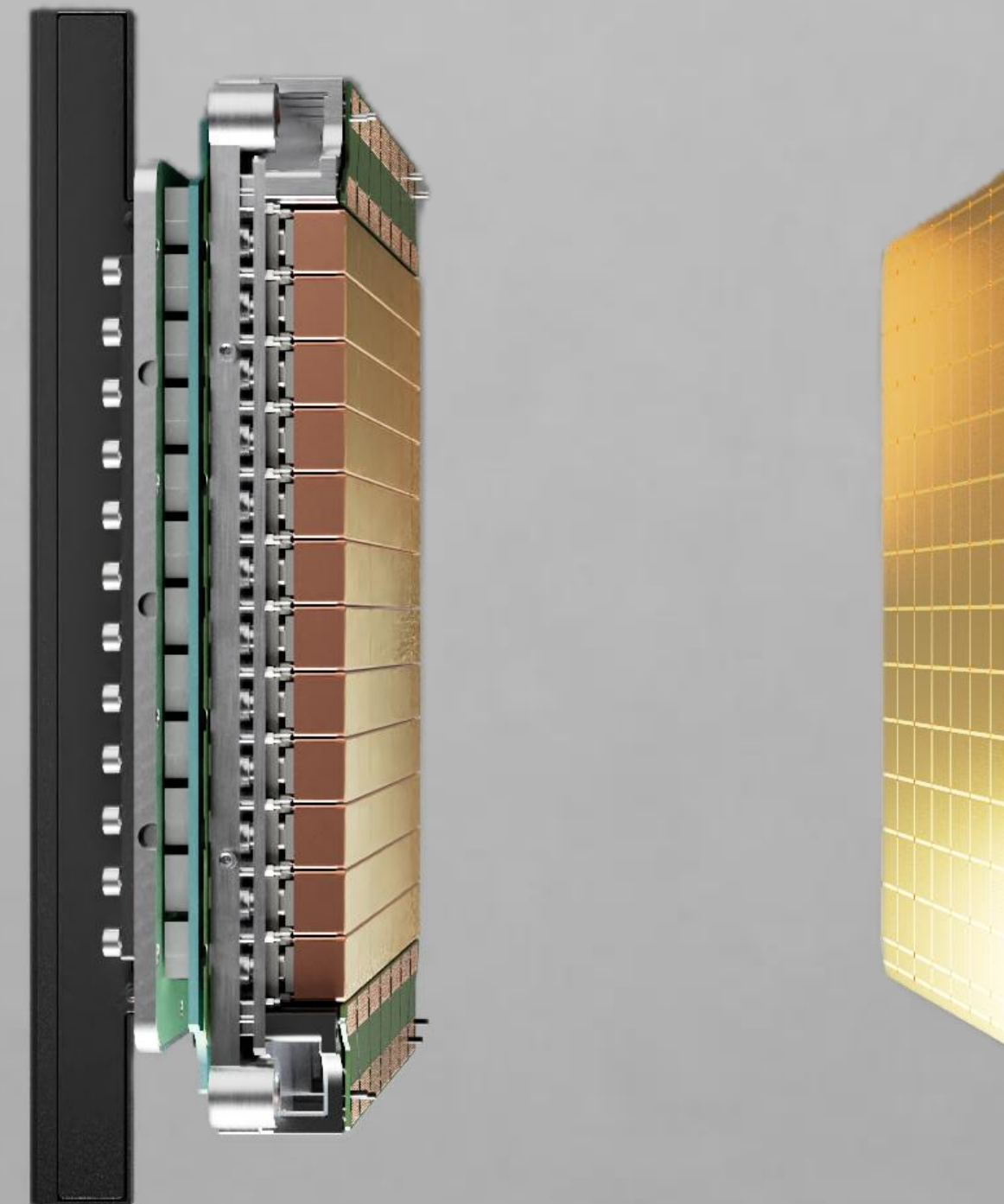
3M

TPS / MW

WAFER-SCALE BUILT OUR 3D INTEGRATION ADVANTAGE

INDUSTRY LEADING
PRODUCTION PROVEN

YIELD-RESILIENT ARCHITECTURE
VERTICAL POWER DELIVERY
INTEGRATED COOLING



CS-6: CEREBRAS WAFER-SCALE SRAM, WITH 3D STACKED DRAM

3D WAFER-SCALE

WORLD'S FASTEST INFERENCE SPEEDS
ORDER OF MAGNITUDE SMALLER FOOTPRINT

WAFER-SCALE SRAM SCALES

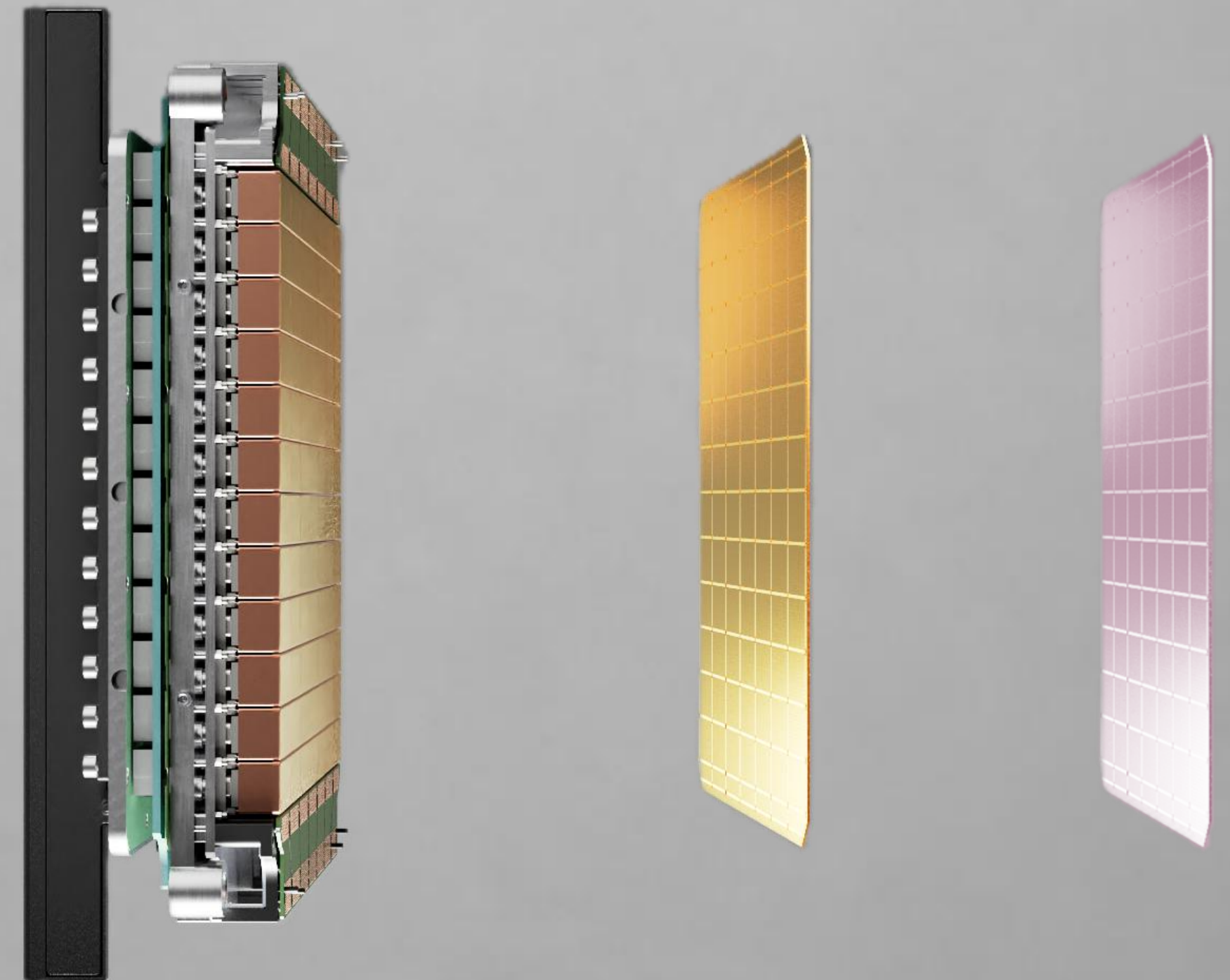
TO THE LARGEST MODELS

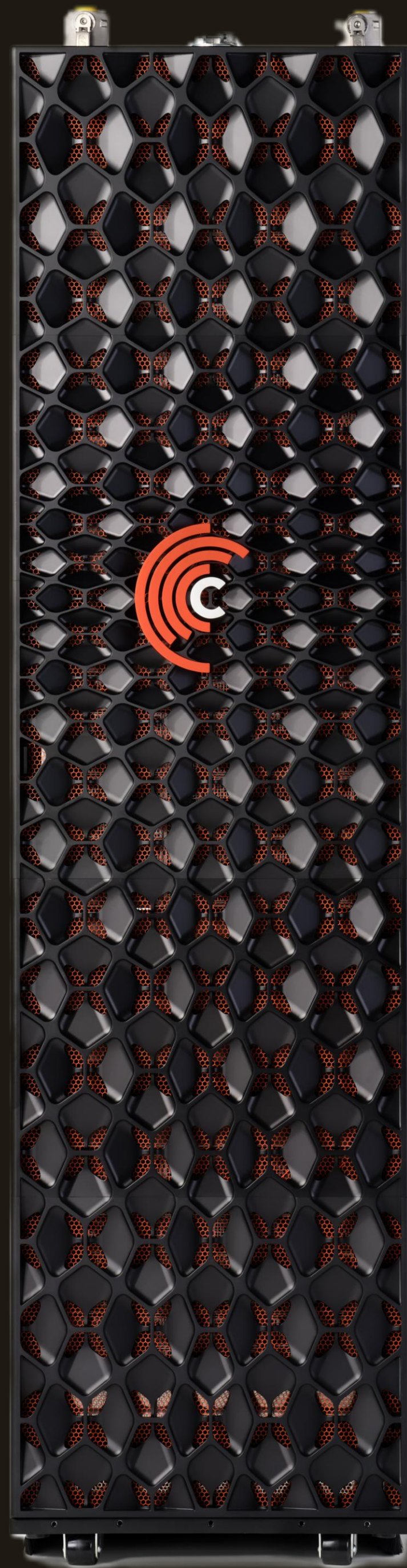
DRAM ENABLES SMALLER FOOTPRINT

LEVERAGING A DECADE OF

3D INTEGRATION INNOVATION

TO BRING ULTRAFAST INFERENCE TO EVERYONE





CS-4

CS-5

CS-6

THE FASTEST GETS FASTER

